Module 3
Linear Transformations

- Linear Transformations: Definition and Examples
- Properties of Linear Transformations
- Linear Transformations - Range and Kernel
- Invertible Linear Transformations
- Matrices of Linear Transformations
- Vector space of linear transformations
- Change of bases
- Similarity
- Application in computer graphics.

Learning objectives

- We will have an understanding of linear transformations and their properties.
- The primary focus will be on matrices as linear transformations between the Euclidean spaces and use the ideas to discuss analogous results for finite dimensional vector spaces.
- We will discuss an application of linear transformation in computer graphics with a real world example.

1 Definition and Examples

We start with the formal definition of a linear transformation and then discuss several examples. Note how the definition is used to prove or disprove that a given function is a linear transformation.

Definition of a linear transformation

Definition 1. Let $\mathcal{V}$ and $\mathcal{W}$ be vector spaces. The mapping $T : \mathcal{V} \to \mathcal{W}$ is called a *linear transformation* if and only if

$$T(cu + v) = cT(u) + T(v)$$

for every choice of u and v in $\mathcal{V}$ and scalars $c \in \mathbb{R}$. In the case for which $\mathcal{V} = \mathcal{W}$, then T is called a *linear operator*.

Example 1. Define a mapping $T : \mathbb{R}^2 \to \mathbb{R}^2$ by

$$T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x + y \\ x - y \end{pmatrix}.$$

(a) Find the images of the coordinate vectors e_1, e_2.

(b) Describe all vectors mapped to the zero vector.

(c) Show T preserves addition and scalar multiplication.

Solution

$$e_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad e_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$T(e_1) = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \qquad T(e_2) = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

Kernel condition:

$$T(x, y) = (0, 0) \Rightarrow x + y = 0,\ x - y = 0 \Rightarrow x = y = 0.$$

Linearity:

$$T(u + v) = T(u) + T(v), \qquad T(cv) = cT(v).$$

Hence T is linear.

Example 2. Define a mapping $T : \mathbb{R}^2 \to \mathbb{R}^2$ by

$$T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x + y \\ x - y \end{pmatrix}.$$

(a) Find the images of the standard basis vectors.

(b) Find all vectors mapped into the zero vector.

(c) Show that T is a linear transformation.

Solution

(a) Images of the standard basis vectors

The standard basis for $\mathbb{R}^2$ is

$$e_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad e_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Applying T,

$$T(e_1) = T\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

$$T(e_2) = T\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

(b) Vectors mapped to the zero vector

We solve

$$T(x, y) = (0, 0).$$

Thus,

$$x + y = 0, \qquad x - y = 0.$$

Adding the equations gives

$$2x = 0 \Rightarrow x = 0.$$

Substituting into the first equation,

$$y = 0.$$

Hence,

$$\ker(T) = \{(0, 0)\}.$$

(c) Verification of linearity

Let

$$u = \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}, \qquad v = \begin{pmatrix} x_2 \\ y_2 \end{pmatrix}.$$

Then

$$u + v = \begin{pmatrix} x_1 + x_2 \\ y_1 + y_2 \end{pmatrix}.$$

Applying T,

$$T(u + v) = \begin{pmatrix} (x_1 + x_2) + (y_1 + y_2) \\ (x_1 + x_2) - (y_1 + y_2) \end{pmatrix}.$$

$$= \begin{pmatrix} x_1 + y_1 \\ x_1 - y_1 \end{pmatrix} + \begin{pmatrix} x_2 + y_2 \\ x_2 - y_2 \end{pmatrix}.$$

$$= T(u) + T(v).$$

Now for scalar c,

$$T(cu) = T\begin{pmatrix} cx_1 \\ cy_1 \end{pmatrix} = \begin{pmatrix} c(x_1 + y_1) \\ c(x_1 - y_1) \end{pmatrix} = cT(u).$$

Therefore,

$$T(cu + v) = cT(u) + T(v).$$

Hence T is a linear transformation.

Example 3. Let A be an $m \times n$ matrix and define

$$T(x) = Ax.$$

(a) Show T is linear.

(b) Compute images of vectors.

Solution

$$A(cu + v) = cAu + Av$$

Hence,

$$T(cu + v) = cT(u) + T(v).$$

Example computation:

$$T(x) = Ax.$$

(Matrix multiplication gives required images.)

Example 4. Let A be an $m \times n$ matrix. Define a mapping

$$T : \mathbb{R}^n \to \mathbb{R}^m$$

by

$$T(\mathbf{x}) = A\mathbf{x}.$$

Show that T is a linear transformation.

Solution

Let

$$\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$$

and let c be a scalar.

We verify the two properties of linearity.

(i) Preservation of addition

Consider

$$T(\mathbf{u} + \mathbf{v}).$$

By definition of T,

$$T(\mathbf{u} + \mathbf{v}) = A(\mathbf{u} + \mathbf{v}).$$

Using properties of matrix multiplication,

$$A(\mathbf{u} + \mathbf{v}) = A\mathbf{u} + A\mathbf{v}.$$

Hence,

$$T(\mathbf{u}+\mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v}).$$

(ii) Preservation of scalar multiplication

Consider

$$T(c\mathbf{u}).$$

Then

$$T(c\mathbf{u}) = A(c\mathbf{u}).$$

Using scalar multiplication property,

$$A(c\mathbf{u}) = c(A\mathbf{u}).$$

Therefore,

$$T(c\mathbf{u}) = cT(\mathbf{u}).$$

Since both properties hold,

$$T(c\mathbf{u}+\mathbf{v}) = cT(\mathbf{u}) + T(\mathbf{v}).$$

Hence $T(\mathbf{x}) = A\mathbf{x}$ is a linear transformation.

Example 5. Define $T : P_3 \to P_2$ by

$$T(p(x)) = p'(x).$$

where $p'(x)$ is the derivative of $p(x)$.

(a) Show that T is a linear transformation.

(b) Find the image of the polynomial $p(x) = 3x^3 + 2x^2 - x + 2$.

(c) Describe the polynomials in P_3 that are mapped to the zero vector of P_2.

Solution

Let

$$p(x) = ax^3 + bx^2 + cx + d.$$

Then

$$T(p) = 3ax^2 + 2bx + c.$$

Derivative properties give

$$T(p+q) = T(p) + T(q), \quad T(cp) = cT(p).$$

Hence linear.

Image:

$$T(p) = 9x^2 + 4x - 1.$$

Kernel: The only functions in P_3 with derivative equal to zero are the constant polynomials $p(x) = c$, where c is a real number, that is,

$$p'(x) = 0 \Rightarrow p(x) = \text{constant}.$$

Proposition 1.1. *Let V and W be vector spaces, and let $T : V \to W$ be a linear transformation. Then $T(0) = 0$.*

Example 6. Define

$$T : \mathbb{R}^2 \to \mathbb{R}^2, \quad T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} e^x \\ e^y \end{pmatrix}.$$

Determine whether T is a linear transformation.

Solution

$$T\begin{pmatrix} 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \neq \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Hence T is NOT linear.

Example 7. Determine whether the function $T : \mathbb{R}^2 \to \mathbb{R}^2$ is a linear transformation.

1. $T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} x + y \\ x - y + 2 \end{bmatrix}$
2. $T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} x \\ y^2 \end{bmatrix}$

Solution

1. $$T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x + y \\ x - y + 2 \end{pmatrix}.$$

   Check zero vector:

   $$T\begin{pmatrix} 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 2 \end{pmatrix} \neq \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

   Fails $T(0) = 0$.

   $$\boxed{\text{T is Not linear}}$$

2. $$T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y^2 \end{pmatrix}.$$

   Check homogeneity:

   $$T\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

   But

   $$T\begin{pmatrix} 0 \\ 2 \end{pmatrix} = \begin{pmatrix} 0 \\ 4 \end{pmatrix} \neq 2\begin{pmatrix} 0 \\ 1 \end{pmatrix} = 2T\begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

   Thus not linear.

   $$\boxed{\text{T is Not linear}}$$

2 Operations with Linear Transformations

Linear transformations can be combined by using a natural addition and scalar multiplication to produce new linear transformations. Let

$$S, T : \mathbb{R}^2 \to \mathbb{R}^2$$

be defined by

$$S\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x+y \\ -x \end{pmatrix} \qquad \text{and} \qquad T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2x-y \\ x+3y \end{pmatrix}.$$

We then define

$$(S+T)(v) = S(v) + T(v) \qquad \text{and} \qquad (cS)(v) = c(S(v)).$$

Example 8. Let

$$v = \begin{pmatrix} 2 \\ -1 \end{pmatrix}; \quad \text{then}$$

$$(S+T)(v) = S(v) + T(v) = \begin{pmatrix} 2+(-1) \\ -2 \end{pmatrix} + \begin{pmatrix} 2(2)-(-1) \\ 2+3(-1) \end{pmatrix} = \begin{pmatrix} 6 \\ -3 \end{pmatrix}.$$

For scalar multiplication let $c = 3$. Then

$$(3T)(v) = 3T(v) = 3\begin{pmatrix} 5 \\ -1 \end{pmatrix} = \begin{pmatrix} 15 \\ -3 \end{pmatrix}.$$

We give the formal definition below. Note the similarity between the operations defined below and the operations we used while discussing vector spaces.

Operations with Linear Transformations

Definition 2. Let V and W be vector spaces and let $S, T : V \to W$ be linear transformations. The function $S+T$ defined by

$$(S+T)(v) = S(v) + T(v)$$

is a linear transformation from V into W. If c is any scalar, the function cS defined by

$$(cS)(v) = cS(v)$$

is a linear transformation from V into W.

It is a good exercise to prove that the set of linear transformations is a vector space under the operations of addition and scaling defined above. This gives us another example of a set which behaves like vectors in $\mathbb{R}^2$ or $\mathbb{R}^3$ spaces, though the geometry is difficult to visualize.

Let us consolidate whatever we have discussed so far.

Fact Summary

Let V, W, and Z be vector spaces and S and T functions from V into W.

1. The function T is a linear transformation provided that for all $\mathbf{u}, \mathbf{v} \in V$ and all scalars c,
$$T(c\mathbf{u} + \mathbf{v}) = cT(\mathbf{u}) + T(\mathbf{v}).$$

2. If A is an $m \times n$ matrix and T is defined by $T(\mathbf{x}) = A\mathbf{x}$, then T is a linear transformation from $\mathbb{R}^n$ into $\mathbb{R}^m$.

3. If T is a linear transformation, then the zero vector in V is mapped to the zero vector in W, that is,
$$T(\mathbf{0}) = \mathbf{0}.$$

4. If $B = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ is an ordered basis for V and $W = \mathbb{R}^n$, then the coordinate mapping
$$T(\mathbf{v}) = [\mathbf{v}]_B$$
is a linear transformation.

5. If $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ is a set of vectors in V and T is a linear transformation, then
$$T(c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_n\mathbf{v}_n) = c_1T(\mathbf{v}_1) + c_2T(\mathbf{v}_2) + \cdots + c_nT(\mathbf{v}_n)$$
for all scalars $c_1, \ldots, c_n$.

6. If S and T are linear transformations and c is a scalar, then $S + T$ and cT are linear transformations.

7. If $T : V \to W$ is a linear transformation and $L : W \to Z$ is a linear transformation, then $L \circ T : V \to Z$ is a linear transformation.

3 Null Space and Range

In this section, we discuss the null space of an $m \times n$ matrix to be the subspace of $\mathbb{R}^n$ of all vectors x with $A\mathbf{x} = 0$. We also defined the column space of A as the subspace of $\mathbb{R}^m$ of all linear combinations of the column vectors of A. In this section we extend these ideas to linear transformations.

Definition of Null Space

Definition 3. Let V and W be vector spaces. For a linear transformation $T : V \to W$ the null space or kernel of T, denoted by $N(T)$ or $Ker(T)$, is defined by

$$N(T) = \{\mathbf{v} \in V \mid T(\mathbf{v}) = 0\}.$$

Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be the linear transformation $T(x) = Ax$. Then the kernel of T is equal to the solution space (Null space) of $Ax = 0$.

Definition of Range

Definition 4. Let V and W be vector spaces. For a linear transformation $T : V \to W$, the range of T, denoted by $R(T)$, is defined by

$$R(T) = \{T(\mathbf{v}) \mid \mathbf{v} \in V\}.$$

Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be the linear transformation $T(x) = Ax$. Then the column space of A is equal to the range of T.

Note 1. The null space of a linear transformation is then the set of all vectors in V that are mapped to the zero vector, with the range being the set of all images of the mapping,

Theorem 3.1. *Let V and W be vector spaces and $T : V \to W$ a linear transformation.*

1. *The null space of T is a subspace of V.*
2. *The range of T is a subspace of W.*

Theorem 3.2. *Let V and W be finite dimensional vector spaces. If $T : V \to W$ is a linear transformation, then*

$$dim(V) = dim(R(T)) + dim(N(T)).$$

Theorem 3.3. *Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be the linear transformation $T(x) = Ax$, then*

$$rank(A) + nullity(A) = n$$

Theorem 3.4. *Let A be an $n \times n$ matrix. Then the following statements are equivalent.*

1. *The matrix A is invertible.*
2. *The linear system $Ax = b$ has a unique solution for every vector b.*
3. *The homogeneous linear system $Ax = 0$ has only the trivial solution.*

4. *The matrix A is row equivalent to the identity matrix.*
5. *The determinant of the matrix A is nonzero.*
6. *The column vectors of A are linearly independent.*
7. *The column vectors of A span $\mathbb{R}^n$.*
8. *The column vectors of A are a basis for $\mathbb{R}^n$.*
9. $\text{rank}(A) = n$.
10. $R(A) = \text{col}(A) = \mathbb{R}^n$.
11. $N(A) = \{0\}$.
12. $\text{row}(A) = \mathbb{R}^n$.
13. *The number of pivot columns of the reduced row echelon form of A is n.*

Let us consolidate.

Fact Summary

Let V and W be vector spaces and T a linear transformation from V into W.

1. The null space $N(T)$ is a subspace of V, and the range $R(T)$ is a subspace of W.
2. If $B = \{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ is a basis for V, then
$$R(T) = \text{span}\{T(\mathbf{v}_1), \ldots, T(\mathbf{v}_n)\}.$$
3. If V and W are finite dimensional vector spaces, then
$$\dim(V) = \dim(R(T)) + \dim(N(T)).$$
4. If A is an $m \times n$ matrix, then
$$\text{rank}(A) + \text{nullity}(A) = n.$$
5. If A is an $m \times n$ matrix, then the rank of A is the number of leading 1s in the row-reduced form of A.
6. If A is an $n \times n$ invertible matrix, we know that
$$\text{rank}(A) = n, \quad R(A) = \text{col}(A) = \mathbb{R}^n, \quad N(A) = \{0\},$$
and the number of leading 1s in the row echelon form of A is n.

Example 9. Define the linear transformation $T : \mathbb{R}^4 \to \mathbb{R}^3$ by

$$T\begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix} = \begin{pmatrix} a+b \\ b-c \\ a+d \end{pmatrix}.$$

(a) Find a basis for the null space of T and its dimension.

(b) Give a description of the range of T.

(c) Find a basis for the range of T and its dimension.

Solution

(a) The null space of T is found by setting each component of the image vector equal to 0. This yields the linear system

$$\begin{cases} a+b=0 \\ b-c=0 \\ a+d=0 \end{cases}$$

This linear system has infinitely many solutions, given by

$$S = \left\{ \left. \begin{pmatrix} -t \\ t \\ t \\ t \end{pmatrix} \right| t \in \mathbb{R} \right\}.$$

Hence,

$$N(T) = \operatorname{span}\left\{ \begin{pmatrix} -1 \\ 1 \\ 1 \\ 1 \end{pmatrix} \right\}.$$

A basis for $N(T)$ consists of the one vector

$$\begin{pmatrix} -1 \\ 1 \\ 1 \\ 1 \end{pmatrix}.$$

Consequently, $\dim(N(T)) = 1$.

(b) Observe that any vector in the range can be written as

$$a\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + b\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + c\begin{pmatrix} 0 \\ -1 \\ 0 \end{pmatrix} + d\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

for some real numbers a, b, c, d. Therefore,

$$R(T) = \operatorname{span}\left\{ \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ -1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\}.$$

(c) Since the range is a subspace of $\mathbb{R}^3$, it has dimension less than or equal to 3. Reducing the matrix

$$\begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix}$$

to

$$\begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & -1 \end{pmatrix},$$

we see pivots in the first three columns.

Thus a basis for $R(T)$ is

$$B = \left\{ \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ -1 \\ 0 \end{pmatrix} \right\}.$$

Therefore $\dim(R(T)) = 3$ and $R(T) = \mathbb{R}^3$.

Example 10. Define $T : \mathbb{R}^3 \to \mathbb{R}^3$ by

$$T \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x - y + 2z \\ 2x + 3y - z \\ -x + 2y - 2z \end{pmatrix}.$$

(a) Find all vectors in $\mathbb{R}^3$ that are mapped to the zero vector.

Solution: The standard matrix of T is

$$A = \begin{pmatrix} 1 & -1 & 2 \\ 2 & 3 & -1 \\ -1 & 2 & -2 \end{pmatrix}.$$

Row reduction of A yields the identity matrix (rank 3). Therefore, $\ker(T) = \{\mathbf{0}\}$. Only the zero vector is mapped to the zero vector.

(b) Let $\mathbf{w} = \begin{pmatrix} 7 \\ -6 \\ -9 \end{pmatrix}$. Determine whether there is a vector $\mathbf{v} \in \mathbb{R}^3$ such that $T(\mathbf{v}) = \mathbf{w}$.

Solution: The augmented matrix $[A \mid \mathbf{w}]$ row-reduces to

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 2 \end{array}\right).$$

The system is consistent with unique solution $\mathbf{v} = \begin{pmatrix} 1 \\ -2 \\ 2 \end{pmatrix}$. Yes, such a vector exists.

Example 11. Suppose that $T : P_2 \to P_2$ is a linear operator such that

$$T(x^2) = 2x - 1, \quad T(-3x) = x^2 - 1, \quad T(-x^2 + 3x) = 2x^2 - 2x + 1.$$

(a) Is it possible to determine $T(2x^2 - 3x + 2)$? If so, find it; and if not, explain why.

Solution: Let $u_1 = x^2$, $u_2 = -3x$, $u_3 = -x^2 + 3x$. Note that $u_1 + u_2 + u_3 = 0$, so $\{u_1, u_2, u_3\}$ is linearly dependent. Any linear combination $au_1 + bu_2 + cu_3$ has zero constant term. However, $2x^2 - 3x + 2$ has constant term $+2$, so it cannot be expressed as such a combination. The system is inconsistent. Therefore, it is **not possible** to determine $T(2x^2 - 3x + 2)$.

(b) Is it possible to determine $T(3x^2 - 4x)$? If so, find it; and if not, explain why.

Solution: $3x^2 - 4x$ has no constant term, so it lies in $\text{span}\{u_1, u_2, u_3\}$. However, the representation is not unique due to the dependence relation. Moreover,

$$T(u_1) + T(u_2) + T(u_3) = (2x - 1) + (x^2 - 1) + (2x^2 - 2x + 1) = 3x^2 - 1 \neq 0.$$

Different representations yield different values for $T(3x^2 - 4x)$. Hence, it is **not possible** to determine a unique value.

Example 12. Find a basis for the null space of the linear transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$,

$$T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 3x + y \\ y \end{pmatrix}.$$

Solution: The matrix is $A = \begin{pmatrix} 3 & 1 \\ 0 & 1 \end{pmatrix}$, which row-reduces to the identity matrix (rank 2). Thus, $\ker(T) = \{\mathbf{0}\}$. The null space has no non-trivial basis (empty basis).

Find a basis for the null space of the linear transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$,

$$T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -x + y \\ x - y \end{pmatrix}.$$

Solution: The matrix is $A = \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix}$. The columns are linearly dependent (second column = -1 times first). Rank is 1. Solving $A\mathbf{v} = \mathbf{0}$ gives the basis

$$\left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\}.$$

Example 13. Find a basis for the range of the linear transformation $T : \mathbb{R}^3 \to \mathbb{R}^3$,

$$T(\mathbf{v}) = \begin{pmatrix} 1 & 1 & 2 \\ 0 & 1 & -1 \\ 2 & 0 & 1 \end{pmatrix} \mathbf{v}.$$

Solution: The columns of the matrix are linearly independent (rank 3). A basis for $R(T)$ is the set of columns:

$$\left\{ \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} \right\}.$$

(Equivalently, $R(T) = \mathbb{R}^3$.)

Example 14. Find a basis for the range of the linear transformation $T : \mathbb{R}^5 \to \mathbb{R}^3$,

$$T(\mathbf{v}) = \begin{pmatrix} 1 & -2 & -3 & 1 & 5 \\ 3 & -1 & 1 & 0 & 4 \\ 1 & 1 & 3 & 1 & 2 \end{pmatrix} \mathbf{v}.$$

Solution: The column space has dimension 3. One basis for $R(T)$ consists of columns 1, 2, and 4:

$$\left\{ \begin{pmatrix} 1 \\ 3 \\ 1 \end{pmatrix}, \begin{pmatrix} -2 \\ -1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \right\}.$$

Example 15. Find a basis for the range of the linear transformation $T : \mathbb{R}^3 \to \mathbb{R}^3$,

$$T \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x \\ y \\ 0 \end{pmatrix}.$$

Solution: The image consists of all vectors with third component zero. A basis for $R(T)$ is

$$\left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \right\}.$$

Example 16. Find a basis for the range of the linear transformation $T : \mathbb{R}^3 \to \mathbb{R}^3$,

$$T \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x - y + 3z \\ x + y + z \\ -x + 3y - 5z \end{pmatrix}.$$

Solution: The matrix has rank 2. A basis for $R(T)$ is given by the first two columns (which are independent):

$$\left\{ \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}, \begin{pmatrix} -1 \\ 1 \\ 3 \end{pmatrix} \right\}.$$

Example 17. If $T : P_2 \to P_2$ is a linear operator and

$$T(1) = 1 + x, \quad T(x) = 2 + x^2, \quad T(x^2) = x - 3x^2,$$

then find $T(-3 + x - x^2)$.

Solution: Since $\{1, x, x^2\}$ is a basis,

$$T(-3 + x - x^2) = -3T(1) + T(x) - T(x^2).$$

Substitute the values:

$$-3(1 + x) + (2 + x^2) - (x - 3x^2) = -3 - 3x + 2 + x^2 - x + 3x^2 = -1 - 4x + 4x^2.$$

Thus, $T(-3 + x - x^2) = -1 - 4x + 4x^2$.

Example 18. Let $T : \mathbb{R}^3 \to \mathbb{R}^3$ be a linear operator and $B = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ a basis for $\mathbb{R}^3$. Suppose

$$T(\mathbf{v}_1) = \begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix}, \quad T(\mathbf{v}_2) = \begin{pmatrix} 0 \\ 5 \\ 0 \end{pmatrix}, \quad T(\mathbf{v}_3) = \begin{pmatrix} -1 \\ -1 \\ 2 \end{pmatrix}.$$

(a) Determine whether

$$\mathbf{w} = \begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix}$$

is in the range of T.

Solution: Form the matrix P with columns $T(\mathbf{v}_1)$, $T(\mathbf{v}_2)$, $T(\mathbf{v}_3)$:

$$P = \begin{pmatrix} -1 & 0 & -1 \\ 2 & 5 & -1 \\ 1 & 0 & 2 \end{pmatrix}.$$

The augmented matrix $[P \mid \mathbf{w}]$ is consistent (row reduction yields a solution, e.g., coefficients $2, -3/5, 0$). Therefore, **yes**, $\mathbf{w} \in R(T)$.

(b) Find a basis for $R(T)$.

Solution: The columns of P are linearly independent (rank 3). A basis for $R(T)$ is

$$\left\{ \begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 5 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ -1 \\ 2 \end{pmatrix} \right\}.$$

(c) Find $\dim(N(T))$.

Solution: By the Rank-Nullity Theorem, $\dim(N(T)) = 3 - rank(P) = 3 - 3 = 0$.

4 Invertible linear transformation: One-to-One and Onto

Definition of a 1-1 and Onto

Definition 5. Let V and W be vector spaces and $T : V \to W$ a mapping.

1. T is called **one-to-one** (injective) if

$$u \neq v \implies T(u) \neq T(v).$$

Equivalently,

$$T(u) = T(v) \Rightarrow u = v.$$

2. T is called **onto** (surjective) if

$$T(V) = W,$$

that is, the range of T equals W.

A mapping that is both one-to-one and onto is called **bijective**.

Note 2. To show T is one-to-one, it suffices to show:

$$T(u) = T(v) \Rightarrow u = v.$$

To show T is onto, we must show:

$$\forall w \in W,\ \exists v \in V \text{ such that } T(v) = w.$$

Example 19. Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be defined by

$$T(\mathbf{v}) = A\mathbf{v}, \quad A = \begin{pmatrix} 1 & 1 \\ -1 & 0 \end{pmatrix}.$$

Show that T is one-to-one and onto.

Solution

Let

$$\mathbf{u} = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}, \quad \mathbf{v} = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}.$$

Then

$$T(\mathbf{u}) = \begin{pmatrix} 1 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{pmatrix} u_1 + u_2 \\ -u_1 \end{pmatrix},$$

and

$$T(\mathbf{v}) = \begin{pmatrix} v_1 + v_2 \\ -v_1 \end{pmatrix}.$$

To show T is one-one:

If $T(\mathbf{u}) = T(\mathbf{v})$, then

$$\begin{pmatrix} u_1 + u_2 \\ -u_1 \end{pmatrix} = \begin{pmatrix} v_1 + v_2 \\ -v_1 \end{pmatrix}.$$

Equating components:

$$-u_1 = -v_1 \Rightarrow u_1 = v_1.$$

Substitute into first equation:

$$u_1 + u_2 = v_1 + v_2 \Rightarrow u_2 = v_2.$$

Thus $\mathbf{u} = \mathbf{v}$, so T is one-to-one.

To show T is onto:

Let

$$\mathbf{w} = \begin{pmatrix} a \\ b \end{pmatrix}.$$

Solve

$$A\mathbf{v} = \mathbf{w}.$$

Compute A^{-1}:

$$A^{-1} = \begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix}.$$

Then

$$\mathbf{v} = A^{-1}\mathbf{w} = \begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} -b \\ a+b \end{pmatrix}.$$

Thus a preimage exists for every $\mathbf{w}$, so T is onto.

Theorem 4.1. *A linear transformation $T : V \to W$ is one-to-one if and only if*

$$N(T) = \{\mathbf{0}\}.$$

Example 20. Define $T : \mathbb{R}^2 \to \mathbb{R}^2$ by

$$T \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2x - 3y \\ 5x + 2y \end{pmatrix}.$$

Show that T is one-to-one.

Solution

$$T \begin{pmatrix} x \\ y \end{pmatrix} = \mathbf{0}$$

if and only if

$$\begin{cases} 2x - 3y = 0 \\ 5x + 2y = 0 \end{cases}$$

By row operation, $x = 0, y = 0$. Thus the only solution is $(0, 0)$. Hence

$$N(T) = \{0\}.$$

By Theorem, T is one-to-one.

Example 21. Let $T : \mathbb{R}^2 \to \mathbb{R}^2$

$$T \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \frac{1}{2}x + \frac{1}{2}y \\ \frac{1}{2}x + \frac{1}{2}y \end{pmatrix}$$

Solve $T(x, y) = 0$:

$$\frac{1}{2}(x + y) = 0$$

$$x + y = 0$$

Let $y = -x$.
Kernel vectors:

$$(x, -x) = x(1, -1)$$

Nontrivial solutions exist.

$$\boxed{T \text{ is NOT one-to-one}}$$

Example 22. Let $T : \mathbb{R}^3 \to \mathbb{R}^3$

$$T\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x + y - z \\ y \\ y - z \end{pmatrix}$$

Solve $T(x, y, z) = 0$:

$$x + y - z = 0$$
$$y = 0$$
$$y - z = 0$$

From $y = 0$:

$$z = 0$$

First equation:

$$x = 0$$

Only trivial solution.

$$\boxed{T \text{ is one-to-one}}$$

Note 3. A linear transformation $T : \mathbb{R}^n \to \mathbb{R}^n$ is onto if and only if its matrix has full rank. For square matrices, this is equivalent to $\det(A) \neq 0$.

Example 23. $T : \mathbb{R}^2 \to \mathbb{R}^2$

$$T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 3x - y \\ x + y \end{pmatrix}$$

Matrix of T:

$$A = \begin{pmatrix} 3 & -1 \\ 1 & 1 \end{pmatrix}$$

Compute determinant:

$$\det(A) = 3(1) - (-1)(1)$$
$$= 3 + 1 = 4 \neq 0$$

$$\boxed{T \text{ is onto}}$$

Example 24. $T : \mathbb{R}^2 \to \mathbb{R}^2$

$$T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -2x + y \\ x - \frac{1}{2}y \end{pmatrix}$$

Matrix:

$$A = \begin{pmatrix} -2 & 1 \\ 1 & -\frac{1}{2} \end{pmatrix}$$

Determinant:

$$\det(A) = (-2)\left(-\frac{1}{2}\right) - (1)(1)$$
$$= 1 - 1 = 0$$

T is NOT onto

Theorem 4.2. *If $T : V \to W$ is linear and $B = \{v_1, \ldots, v_n\}$ is a basis for V, and T is one-to-one, then*

$$\{T(v_1), \ldots, T(v_n)\}$$

is a basis for $R(T)$.

Definition of an isomorphism

Definition 6. A linear transformation $T : V \to W$ that is both one-to-one and onto is called an **isomorphism**. In this case, V and W are said to be **isomorphic**.

Theorem 4.3. *If V is a vector space with $\dim(V) = n$, then $V \cong \mathbb{R}^n$.*

Definition of an Inverse of a Linear Transformation

Definition 7. Let $T : V \to W$ be a one-to-one linear transformation. The mapping

$$T^{-1} : R(T) \to V$$

defined by

$$T^{-1}(w) = v \quad \text{if and only if} \quad T(v) = w$$

is called the **inverse of** T.
If T is onto, then T^{-1} is defined on all of W.

Proposition 4.1. *Let V and W be vector spaces and $T : V \to W$ a one-to-one linear transformation. Then the mapping $T^{-1} : R(T) \to V$ is also a linear transformation.*

Proposition 4.2. *Let A be an $n \times n$ invertible matrix and $T : \mathbb{R}^n \to \mathbb{R}^n$ the linear transformation defined by*

$$T(\mathbf{x}) = A\mathbf{x}.$$

Then

$$T^{-1}(\mathbf{x}) = A^{-1}\mathbf{x}.$$

Example 25. Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be defined by $T(\mathbf{v}) = A\mathbf{v}$, where

$$A = \begin{bmatrix} 1 & 1 \\ -1 & 0 \end{bmatrix}.$$

Verify that the inverse map $T^{-1} : \mathbb{R}^2 \to \mathbb{R}^2$ is given by $T^{-1}(\mathbf{w}) = A^{-1}\mathbf{w}$, where

$$A^{-1} = \begin{bmatrix} 0 & -1 \\ 1 & 1 \end{bmatrix}.$$

Solution

Let

$$\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \in \mathbb{R}^2.$$

Then

$$\mathbf{w} = T(\mathbf{v}) = A\mathbf{v} = \begin{bmatrix} 1 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} v_1 + v_2 \\ -v_1 \end{bmatrix}.$$

$$A^{-1}\mathbf{w} = \begin{bmatrix} 0 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} v_1 + v_2 \\ -v_1 \end{bmatrix}.$$

$$= \begin{bmatrix} 0(v_1 + v_2) + (-1)(-v_1) \\ 1(v_1 + v_2) + 1(-v_1) \end{bmatrix} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}.$$

$$T^{-1}(\mathbf{w}) = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}.$$

Theorem 4.4. *If V and W are vector spaces of dimension n, then V and W are isomorphic.*

5 Coordinates

Let V be a vector space with $\dim(V) = n$, and $B = \{v_1, v_2, \ldots, v_n\}$ an ordered basis for V. Let

$$T : V \to \mathbb{R}^n$$

be the map that sends a vector v in V to its coordinate vector in $\mathbb{R}^n$ relative to B. That is,

$$T(v) = [v]_B.$$

The coordinate vector of v relative to B is unique. The map T is also a linear transformation.

Example 26. Let $T : \mathbb{R}^3 \to \mathbb{R}^2$ be linear with

$$T(e_1) = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad T(e_2) = \begin{pmatrix} -1 \\ 2 \end{pmatrix}, \quad T(e_3) = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Find $T(v)$ for

$$v = \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix}.$$

Solution

$$v = e_1 + 3e_2 + 2e_3.$$

$$T(v) = T(e_1) + 3T(e_2) + 2T(e_3)$$

$$= \begin{pmatrix} 1 \\ 1 \end{pmatrix} + 3\begin{pmatrix} -1 \\ 2 \end{pmatrix} + 2\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} -2 \\ 9 \end{pmatrix}.$$

Example 27. Let $T : \mathbb{R}^3 \to \mathbb{R}^2$ be a linear transformation, and let B be the standard basis for $\mathbb{R}^3$.

If

$$T(e_1) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad T(e_2) = \begin{bmatrix} -1 \\ 2 \end{bmatrix}, \qquad \text{and} \qquad T(e_3) = \begin{bmatrix} 0 \\ 1 \end{bmatrix},$$

find $T(\mathbf{v})$, where

$$\mathbf{v} = \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix}.$$

Solution

To find the image of the vector $\mathbf{v}$, we first write the vector as a linear combination of the basis vectors. In this case,

$$\mathbf{v} = e_1 + 3e_2 + 2e_3.$$

Applying T to this linear combination and using the linearity properties of T, we have

$$T(\mathbf{v}) = T(e_1 + 3e_2 + 2e_3).$$

$$= T(e_1) + 3T(e_2) + 2T(e_3).$$

Substituting the given values,

$$= \begin{bmatrix} 1 \\ 1 \end{bmatrix} + 3 \begin{bmatrix} -1 \\ 2 \end{bmatrix} + 2 \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

$$= \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \begin{bmatrix} -3 \\ 6 \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \end{bmatrix} = \begin{bmatrix} -2 \\ 9 \end{bmatrix}.$$

$$\boxed{T(\mathbf{v}) = \begin{bmatrix} -2 \\ 9 \end{bmatrix}}$$

Example 28. Let $T : \mathbb{R}^3 \to \mathbb{R}^3$ be linear and

$$B = \{v_1, v_2, v_3\}$$

a basis. Find $T(2, 3, 6)$.

Suppose

$$(2, 3, 6) = c_1 v_1 + c_2 v_2 + c_3 v_3.$$

Solving gives

$$c_1 = -1, \quad c_2 = 1, \quad c_3 = 2.$$

Using linearity,

$$T(2, 3, 6) = -T(v_1) + T(v_2) + 2T(v_3).$$

Example 29. Let $T : \mathbb{R}^3 \to \mathbb{R}^3$ be a linear operator and B a basis for $\mathbb{R}^3$ given by

$$B = \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} \right\}.$$

If

$$T\left(\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right) = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \qquad T\left(\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \right) = \begin{bmatrix} -1 \\ -2 \\ -3 \end{bmatrix}, \qquad T\left(\begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} \right) = \begin{bmatrix} 2 \\ 2 \\ 4 \end{bmatrix}.$$

Find

$$T\left(\begin{bmatrix} 2 \\ 3 \\ 6 \end{bmatrix} \right).$$

Solution

Since B is a basis for $\mathbb{R}^3$, there exist unique scalars c_1, c_2, and c_3 such that

$$c_1 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + c_3 \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ 6 \end{bmatrix}.$$

Solving this system, we obtain

$$c_1 = -1, \qquad c_2 = 1, \qquad c_3 = 2.$$

Hence,

$$\begin{bmatrix} 2 \\ 3 \\ 6 \end{bmatrix} = -1 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + 2 \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}.$$

Therefore,

$$T\left(\begin{bmatrix} 2 \\ 3 \\ 6 \end{bmatrix} \right) = T\left(-1 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + 2 \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} \right).$$

By linearity of T,

$$T\left(\begin{bmatrix} 2 \\ 3 \\ 6 \end{bmatrix} \right) = (-1)T\left(\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right) + T\left(\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \right) + 2T\left(\begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} \right).$$

Substituting given values,

$$= - \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + \begin{bmatrix} -1 \\ -2 \\ -3 \end{bmatrix} + 2 \begin{bmatrix} 2 \\ 2 \\ 4 \end{bmatrix}.$$

$$= \begin{bmatrix} -1 \\ -1 \\ -1 \end{bmatrix} + \begin{bmatrix} -1 \\ -2 \\ -3 \end{bmatrix} + \begin{bmatrix} 4 \\ 4 \\ 8 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 4 \end{bmatrix}.$$

$$\boxed{T\left(\begin{bmatrix} 2 \\ 3 \\ 6 \end{bmatrix} \right) = \begin{bmatrix} 2 \\ 1 \\ 4 \end{bmatrix}}$$

6 Change of Basis

Many problems in applied mathematics are made easier by changing from one basis of a vector space to another. To simplify our explanation of the process, we will consider a vector space V with $\dim(V) = 2$ and show how to change from coordinates relative to one basis for V to another basis for V.

Let V be a vector space of dimension 2 and let

$$B = \{v_1, v_2\} \quad \text{and} \quad B' = \{v_1', v_2'\}$$

be ordered bases for V. Now let v be a vector in V, and suppose that the coordinates of v relative to B are given by

$$[v]_B = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad \text{that is} \quad v = x_1 v_1 + x_2 v_2.$$

To determine the coordinates of v relative to B', we first write v_1 and v_2 in terms of the vectors v_1' and v_2'. Since B' is a basis, there are scalars a_1, a_2, b_1, and b_2 such that

$$v_1 = a_1 v_1' + a_2 v_2',$$

$$v_2 = b_1 v_1' + b_2 v_2'.$$

Then v can be written as

$$v = x_1(a_1 v_1' + a_2 v_2') + x_2(b_1 v_1' + b_2 v_2').$$

Collecting the coefficients of v_1' and v_2' gives

$$v = (x_1 a_1 + x_2 b_1) v_1' + (x_1 a_2 + x_2 b_2) v_2',$$

so that the coordinates of v relative to the basis B' are given by

$$[v]_{B'} = \begin{bmatrix} x_1 a_1 + x_2 b_1 \\ x_1 a_2 + x_2 b_2 \end{bmatrix}.$$

Now by rewriting the vector on the right-hand side as a matrix product, we have

$$[v]_{B'} = \begin{bmatrix} a_1 & b_1 \\ a_2 & b_2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} a_1 & b_1 \\ a_2 & b_2 \end{bmatrix} [v]_B.$$

Notice that the column vectors of the matrix are the coordinate vectors $[v_1]_{B'}$ and $[v_2]_{B'}$. The matrix

$$\begin{bmatrix} a_1 & b_1 \\ a_2 & b_2 \end{bmatrix}$$

is called the **transition matrix** from B to B' and is denoted by $[I]_{B'}^{B}$, so that

$$[v]_{B'} = [I]_{B'}^{B} [v]_B.$$

Example 30. Let $V = \mathbb{R}^2$ with bases

$$B = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\} \quad \text{and} \quad B' = \left\{ \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \end{bmatrix} \right\}.$$

a. Find the transition matrix from B to B'.

b. Let $[v]_B = \begin{bmatrix} 3 \\ -2 \end{bmatrix}$ and find $[v]_{B'}$.

Solution

a. By denoting the vectors in B by v_1 and v_2 and those in B' by v_1' and v_2', the column vectors of the transition matrix are $[v_1]_{B'}$ and $[v_2]_{B'}$. These coordinate vectors are found from the equations

$$c_1 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \text{and} \quad d_1 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + d_2 \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

Solving these equations gives $c_1 = 2$ and $c_2 = 3$, and $d_1 = 0$ and $d_2 = -1$, so that

$$[v_1]_{B'} = \begin{bmatrix} 2 \\ 3 \end{bmatrix} \quad \text{and} \quad [v_2]_{B'} = \begin{bmatrix} 0 \\ -1 \end{bmatrix}.$$

Therefore, the transition matrix is

$$[I]_{B'}^{B} = \begin{bmatrix} 2 & 0 \\ 3 & -1 \end{bmatrix}.$$

b. Since

$$[v]_{B'} = [I]_{B'}^{B} [v]_B,$$

then

$$[v]_{B'} = \begin{bmatrix} 2 & 0 \\ 3 & -1 \end{bmatrix} \begin{bmatrix} 3 \\ -2 \end{bmatrix} = \begin{bmatrix} 6 \\ 11 \end{bmatrix}.$$

Observe that the same vector, relative to the different bases, is obtained from the coordinates $[v]_B$ and $[v]_{B'}$. That is,

$$3 \begin{bmatrix} 1 \\ 1 \end{bmatrix} - 2 \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \end{bmatrix} = 6 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + 11 \begin{bmatrix} -1 \\ 1 \end{bmatrix}.$$

Theorem 6.1. *Let V be a vector space of dimension n with ordered bases*

$$B = \{v_1, v_2, \ldots, v_n\} \quad \textit{and} \quad B' = \{v_1', v_2', \ldots, v_n'\}.$$

Then the transition matrix from B to B' is given by

$$[I]_{B'}^{B} = \Big[[v_1]_{B'} \; [v_2]_{B'} \; \cdots \; [v_n]_{B'} \Big].$$

Moreover, a change of coordinates is carried out by

$$[v]_{B'} = [I]_{B'}^{B} [v]_B.$$

Example 31. Let $V = \mathcal{P}_2$ with bases

$$B = \{1, x, x^2\} \quad \text{and} \quad B' = \{1, x+1, x^2+x+1\}.$$

a. Find the transition matrix $[I]_{B'}^{B}$.

b. Let $p(x) = 3 - x + 2x^2$ and find $[p(x)]_{B'}$.

Solution

a. To find the first column vector of the transition matrix, we must find scalars a_1, a_2, and a_3 such that

$$a_1(1) + a_2(x+1) + a_3(x^2+x+1) = 1.$$

By inspection we see that the solution is $a_1 = 1, a_2 = 0, a_3 = 0$. Therefore,

$$[1]_{B'} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}.$$

The second and third column vectors of the transition matrix can be found by solving the equations

$$b_1(1) + b_2(x+1) + b_3(x^2+x+1) = x$$

and

$$c_1(1) + c_2(x+1) + c_3(x^2+x+1) = x^2$$

respectively. The solutions are given by $b_1 = -1, b_2 = 1, b_3 = 0$, and $c_1 = 0, c_2 = -1, c_3 = 1$. Hence, the transition matrix is

$$[I]_{B'}^{B} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}.$$

b. The basis B is the standard basis for $\mathcal{P}_2$, so the coordinate vector of $p(x) = 3 - x + 2x^2$ relative to B is given by

$$[p(x)]_B = \begin{bmatrix} 3 \\ -1 \\ 2 \end{bmatrix}.$$

Hence,

$$[p(x)]_{B'} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 4 \\ -3 \\ 2 \end{bmatrix}.$$

Notice that $3 - x + 2x^2 = 4(1) - 3(x+1) + 2(x^2+x+1)$.

Example 32. Let $B = \{e_1, e_2\}$ be the standard ordered basis for $\mathbb{R}^2$, B' be the ordered basis given by

$$B' = \{v_1', v_2'\} = \left\{ \begin{bmatrix} -1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\}$$

and let $v = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$.

(a) Find the transition matrix from B to B'.

(b) Find $[v]_{B'}$.

(c) Write the vector v as a linear combination of e_1 and e_2 and also as a linear combination of v_1' and v_2'.

(d) Show the results of part (c) graphically.

Solution

a. The transition matrix from B to B' is computed by solving the equations

$$c_1\begin{bmatrix}-1\\1\end{bmatrix}+c_2\begin{bmatrix}1\\1\end{bmatrix}=\begin{bmatrix}1\\0\end{bmatrix}\quad\text{and}\quad d_1\begin{bmatrix}-1\\1\end{bmatrix}+d_2\begin{bmatrix}1\\1\end{bmatrix}=\begin{bmatrix}0\\1\end{bmatrix}.$$

That is, we must solve the linear systems

$$\begin{cases}-c_1+c_2=1\\c_1+c_2=0\end{cases}\quad\text{and}\quad\begin{cases}-d_1+d_2=0\\d_1+d_2=1\end{cases}.$$

The unique solutions are given by $c_1=-\frac{1}{2}$, $c_2=\frac{1}{2}$ and $d_1=\frac{1}{2}$, $d_2=\frac{1}{2}$. The transition matrix is then given by

$$[I]_{B'}^{B}=\begin{bmatrix}-\frac{1}{2}&\frac{1}{2}\\\frac{1}{2}&\frac{1}{2}\end{bmatrix}.$$

b. Since B is the standard basis, the coordinates of v relative to B are $[v]_B=\begin{bmatrix}3\\4\end{bmatrix}$. By Theorem 14, the coordinates of v relative to B' are given by

$$[v]_{B'}=\begin{bmatrix}-\frac{1}{2}&\frac{1}{2}\\\frac{1}{2}&\frac{1}{2}\end{bmatrix}\begin{bmatrix}3\\4\end{bmatrix}=\begin{bmatrix}\frac{1}{2}\\\frac{1}{2}\end{bmatrix}.$$

c. Using the coordinates of v relative to the two bases, we have

$$3\begin{bmatrix}1\\0\end{bmatrix}+4\begin{bmatrix}0\\1\end{bmatrix}=v=\frac{1}{2}v_1'+\frac{1}{2}v_2'.$$

(Note: the extraction had a minor typo in coefficients; the correct linear combination consistent with the matrix multiplication is $\frac{1}{2}v_1'+\frac{1}{2}v_2'$.)

d. The picture given in Fig. 2 shows the location of the terminal point $(3,4)$ of the vector v relative to the e_1e_2 axes and the $v_1'v_2'$ axes.

7 Inverse of a Transition Matrix

A fact that will be useful in Chap. 4 is that the transition matrix $[I]_{B'}^{B}$ between bases B and B' of a finite dimensional vector space is invertible. Moreover, the transition matrix from B' to B is the inverse of $[I]_{B'}^{B}$. To see this, suppose that V is a vector space of dimension n with ordered bases

$$B=\{v_1,v_2,\ldots,v_n\}\quad\text{and}\quad B'=\{v_1',v_2',\ldots,v_n'\}.$$

To show that $[I]_{B'}^{B}$ is invertible, let $\mathbf{x} \in \mathbb{R}^n$ be such that

$$[I]_{B'}^{B}\mathbf{x} = \mathbf{0}.$$

Observe that the left-hand side of this equation in vector form is $x_1[v_1]_{B'} + \cdots + x_n[v_n]_{B'}$. Since B is a basis, then the vectors v_i for $1 \leq i \leq n$ are linearly independent. Hence, so are the vectors $[v_1]_{B'}, \ldots, [v_n]_{B'}$. Therefore, $x_1 = x_2 = \cdots = x_n = 0$. Since the only solution to the homogeneous equation $[I]_{B'}^{B}\mathbf{x} = \mathbf{0}$ is the trivial solution, then by Theorem 17 of Sec. 1.6, the matrix $[I]_{B'}^{B}$ is invertible. Moreover, by Theorem 14, since

$$[v]_{B'} = [I]_{B'}^{B}[v]_B \quad \text{we know that} \quad ([I]_{B'}^{B})^{-1}[v]_{B'} = [v]_B$$

and therefore

$$[I]_{B}^{B'} = ([I]_{B'}^{B})^{-1}.$$

Theorem 7.1. *Let V be a vector space of dimension n with ordered bases*

$$B = \{v_1, v_2, \ldots, v_n\} \quad \textit{and} \quad B' = \{v_1', v_2', \ldots, v_n'\}.$$

Then the transition matrix $[I]_{B'}^{B}$ from B to B' is invertible and

$$[I]_{B}^{B'} = \left([I]_{B'}^{B}\right)^{-1}.$$

Let us consolidate.

Fact Summary

Let V be a vector space with $\dim(V) = n$.

1. In $\mathbb{R}^n$, the coordinates of a vector with respect to the standard basis are the components of the vector.
2. Given any two ordered bases for V, a transition matrix can be used to change the coordinates of a vector relative to one basis to the coordinates relative to the other basis.
3. If B and B' are two ordered bases for V, the transition matrix from B to B' is the matrix $[I]_{B'}^{B}$ whose column vectors are the coordinates of the basis vectors of B relative to the basis B'. Also
$$[v]_{B'} = [I]_{B'}^{B}[v]_B.$$
4. If B and B' are two ordered bases for V, the transition matrix from B' to B is invertible and the inverse matrix is the transition matrix from B to B'. That is,
$$\left([I]_{B'}^{B}\right)^{-1} = [I]_{B}^{B'}.$$

8 Matrix Representation of a Linear Transformation

Matrices have played an important role in the study of linear algebra. In this section we establish the connection between matrices and linear transformations.

Recall that given any $m \times n$ matrix A, we can define a linear transformation

$$T : \mathbb{R}^n \to \mathbb{R}^m$$

by

$$T(\mathbf{v}) = A\mathbf{v}.$$

A linear transformation $T : \mathbb{R}^3 \to \mathbb{R}^2$ was completely determined by the images of the coordinate vectors $\mathbf{e}_1, \mathbf{e}_2$, and $\mathbf{e}_3$.

Let

$$\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}.$$

Then

$$\mathbf{v} = v_1\mathbf{e}_1 + v_2\mathbf{e}_2 + v_3\mathbf{e}_3,$$

so that

$$T(\mathbf{v}) = v_1T(\mathbf{e}_1) + v_2T(\mathbf{e}_2) + v_3T(\mathbf{e}_3).$$

Example 33. Suppose

$$T(\mathbf{e}_1) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad T(\mathbf{e}_2) = \begin{bmatrix} -1 \\ 2 \end{bmatrix}, \quad T(\mathbf{e}_3) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Let A be the matrix whose columns are $T(\mathbf{e}_1), T(\mathbf{e}_2)$ and $T(\mathbf{e}_3)$:

$$A = \begin{bmatrix} 1 & -1 & 0 \\ 1 & 2 & 1 \end{bmatrix}.$$

Then $T(\mathbf{v}) = A\mathbf{v}$. Thus, every linear transformation can be expressed as a matrix product.

General Case

If $T : \mathbb{R}^n \to \mathbb{R}^m$ is linear, then

$$T(\mathbf{v}) = A\mathbf{v},$$

where the jth column of A is $T(\mathbf{e}_j)$, for $j = 1, 2, \ldots, n$.

The matrix A is called the **matrix representation of T relative to the standard bases**.

9 Matrix Representation Relative to Bases

Let V and W be finite dimensional vector spaces with ordered bases

$$B = \{v_1, v_2, \ldots, v_n\}, \quad B' = \{w_1, w_2, \ldots, w_m\}.$$

Let $T : V \to W$ pbe a linear transformation.

For any vector $v \in V$, define its coordinate vector relative to B by

$$[v]_B = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix}.$$

Hence,

$$v = c_1 v_1 + c_2 v_2 + \cdots + c_n v_n.$$

Applying T gives

$$T(v) = c_1 T(v_1) + c_2 T(v_2) + \cdots + c_n T(v_n).$$

Representation of Images

Each vector $T(v_i)$ belongs to W. Therefore, there exist scalars a_{ij} such that

$$\begin{aligned} T(v_1) &= a_{11} w_1 + a_{21} w_2 + \cdots + a_{m1} w_m, \\ T(v_2) &= a_{12} w_1 + a_{22} w_2 + \cdots + a_{m2} w_m, \\ &\vdots \\ T(v_n) &= a_{1n} w_1 + a_{2n} w_2 + \cdots + a_{mn} w_m. \end{aligned}$$

Thus,

$$[T(v_i)]_{B'} = \begin{bmatrix} a_{1i} \\ a_{2i} \\ \vdots \\ a_{mi} \end{bmatrix}.$$

Matrix Form

Hence,

$$[T(v)]_{B'} = c_1 \begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{bmatrix} + c_2 \begin{bmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{bmatrix} + \cdots + c_n \begin{bmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{bmatrix}.$$

Equivalently,

$$[T(v)]_{B'} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix}.$$

The matrix above is denoted by

$$[T]_B^{B'} = \begin{bmatrix} [T(v_1)]_{B'} & [T(v_2)]_{B'} & \cdots & [T(v_n)]_{B'} \end{bmatrix}.$$

It is called the **matrix of** T **relative to** B **and** B'.

Theorem 9.1. *Let V and W be finite dimensional vector spaces with ordered bases*

$$B = \{v_1, \ldots, v_n\}, \qquad B' = \{w_1, \ldots, w_m\},$$

and let $T : V \to W$ be linear.

Then the matrix $[T]_B^{B'}$ represents T relative to the bases B and B'. Moreover,

$$[T(v)]_{B'} = [T]_B^{B'}[v]_B.$$

Special Case: Change of Basis

If $V = W$, the bases B and B' are different, and T is the identity transformation,

$$T(v) = v,$$

then

$$[T]_B^{B'}$$

is the **change-of-basis matrix** denoted by

$$[I]_B^{B'}.$$

Example 34. Define the linear operator $T : \mathbb{R}^3 \to \mathbb{R}^3$ by

$$T\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x \\ -y \\ z \end{pmatrix}.$$

(a) Find the matrix of T relative to the standard basis.

(b) Use the result to compute $T(1, 1, 2)^T$.

Solution

Let $B = \{e_1, e_2, e_3\}$ be the standard basis.

$$T(e_1) = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad T(e_2) = \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix}, \quad T(e_3) = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

Hence

$$[T]_B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

For

$$v = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix},$$

we obtain

$$T(v) = [T]_B v = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}.$$

Steps for Finding Matrix Representation

1. Compute $T(v_1), \ldots, T(v_n)$.
2. Express each image relative to basis B'.
3. Form matrix $[T]_B^{B'}$ using these coordinate vectors.
4. Compute $[v]_B$.
5. Evaluate
$$[T(v)]_{B'} = [T]_B^{B'}[v]_B.$$
6. Reconstruct
$$T(v) = c_1 w_1 + \cdots + c_m w_m.$$

Example 35. Let $T : \mathbb{R}^2 \to \mathbb{R}^3$ be

$$T(x_1, x_2) = \begin{bmatrix} x_2 \\ x_1 + x_2 \\ x_1 - x_2 \end{bmatrix}.$$

Let

$$B = \left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ 1 \end{bmatrix} \right\}, \quad B' = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\}$$

be ordered bases for $\mathbb{R}^2$ and $\mathbb{R}^3$, respectively.

(a) Find the matrix $[T]_B^{B'}$.

(b) Let $v = \begin{bmatrix} -3 \\ -2 \end{bmatrix}$. Find $T(v)$ directly and then use the matrix found in part (a).

Solution

(a) We first apply T to the basis vectors of B, which gives

$$T \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix}, \qquad T \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \\ 2 \end{bmatrix}.$$

Solving coordinate systems gives

$$[T]_B^{B'} = \begin{bmatrix} -1 & -3 \\ 4 & 2 \\ -1 & 2 \end{bmatrix}.$$

(b) For

$$v = \begin{bmatrix} -3 \\ -2 \end{bmatrix},$$

its coordinates relative to B are

$$[v]_B = \begin{bmatrix} -\frac{3}{5} \\ -\frac{4}{5} \end{bmatrix}.$$

Hence

$$[T(v)]_{B'} = \begin{bmatrix} -1 & -3 \\ 4 & 2 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} -\frac{3}{5} \\ -\frac{4}{5} \end{bmatrix} = \begin{bmatrix} 3 \\ -4 \\ -1 \end{bmatrix}.$$

Therefore

$$T(v) = \begin{bmatrix} -2 \\ -5 \\ -1 \end{bmatrix}.$$

Example 36. Define a linear transformation $T : P_2 \to P_3$ by

$$T(f(x)) = x^2 f''(x) - 2f'(x) + xf(x).$$

Find the matrix representation of T relative to the standard bases for P_2 and P_3.

Standard bases:

$$B = \{1, x, x^2\}, \quad B' = \{1, x, x^2, x^3\}.$$

Solution

$$T(1) = x, \qquad T(x) = x^2 - 2,$$
$$T(x^2) = x^3 + 2x^2 - 4x.$$

Coordinate vectors:

$$[T(1)]_{B'} = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \quad [T(x)]_{B'} = \begin{bmatrix} -2 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \quad [T(x^2)]_{B'} = \begin{bmatrix} 0 \\ -4 \\ 2 \\ 1 \end{bmatrix}.$$

Thus

$$[T]_B^{B'} = \begin{bmatrix} 0 & -2 & 0 \\ 1 & 0 & -4 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}.$$

Theorem 9.2. *Let S and T be linear transformations from V to W. Then*

$$[S+T]_B^{B'} = [S]_B^{B'} + [T]_B^{B'},$$

and for scalar k,

$$[kT]_B^{B'} = k[T]_B^{B'}.$$

Example 37. Let

$$S(x,y) = (x+2y, -y), \quad T(x,y) = (-x+y, 3x).$$

Standard basis B of $\mathbb{R}^2$.

$$[S]_B = \begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix}, \quad [T]_B = \begin{bmatrix} -1 & 1 \\ 3 & 0 \end{bmatrix}.$$

Hence

$$[S+T]_B = \begin{bmatrix} 0 & 3 \\ 3 & -1 \end{bmatrix},$$

and

$$[3S]_B = \begin{bmatrix} 3 & 6 \\ 0 & -3 \end{bmatrix}.$$

Theorem 9.3. *Let*

$$T : U \to V, \qquad S : V \to W$$

be linear transformations.

Then the matrix of the composition satisfies

$$[S \circ T]_B^{B''} = [S]_{B'}^{B''} [T]_B^{B'}.$$

Since $T^{-1}T = I$, we have

$$[I]_B = [T^{-1}T]_B = [T^{-1}]_B [T]_B.$$

Since $[I]_B$ is the identity matrix,

$$[T^{-1}]_B = ([T]_B)^{-1}.$$

10 Similarity

The matrix for a linear operator $T : V \to V$ depends on the basis selected for V. One of the fundamental problems of linear algebra is to choose a basis for V that makes the matrix for T as simple as possible-a diagonal or a triangular matrix, for example. In this section we will study this problem.

Problem. If B and B' are two bases for a finite-dimensional vector space V, and if $T : V \to V$ is a linear operator, what relationship, if any, exists between the matrices $[T]_B$ and $[T]_{B'}$?

Theorem 10.1. *Let $T : V \to V$ be a linear operator on a finite-dimensional vector space V, and let B and B' be bases for V. Then*

$$[T]_{B'} = P^{-1}[T]_B P$$

where $P = P_{B' \to B}$ and $P^{-1} = P_{B \to B'}$.

Theorem 10.2. *If V is a finite-dimensional vector space, then two matrices A and B represent the same linear operator (but possibly with respect to different bases) if and only if they are similar. Moreover, if*

$$B = P^{-1}AP,$$

then P is the transition matrix from the basis used for B to the basis used for A.

11 Application in Computer Graphics

In computer graphics, all visual components—such as points, lines, shapes, 3D models, and camera motions—are modeled using vectors and matrices. Linear transformations provide a powerful framework to:

- Perform geometric operations such as rotation, scaling, shearing, reflection, and translation in a systematic manner.
- Combine multiple transformations into a single operation through matrix multiplication.
- Map three-dimensional scenes onto a two-dimensional screen using projection methods (perspective or orthographic).
- Enable real-time animation through efficient matrix computations carried out by the GPU.

Mathematically, a linear transformation $T : \mathbb{R}^n \to \mathbb{R}^n$ can be represented by an $n \times n$ matrix A such that

$$T(\mathbf{v}) = A\mathbf{v}.$$

In practical applications, particularly in graphics, **homogeneous coordinates** are used to represent affine transformations (including translation) within a unified matrix framework.

Key 2D Transformations (Standard Basis)

1. **Scaling** by factors s_x, s_y:
$$A = \begin{pmatrix} s_x & 0 \\ 0 & s_y \end{pmatrix}$$
2. **Rotation** by angle θ (counter-clockwise):
$$A = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$
3. **Reflection** over x-axis:
$$A = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

4. **Shearing** (parallel to x-axis by factor k):

$$A = \begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}$$

5. **Translation** by (t_x, t_y) (requires homogeneous coordinates):

$$A = \begin{pmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{pmatrix}$$

Composite transformation: Apply matrices from right to left, i.e., $M_3(M_2(M_1\mathbf{v}))$.

Homogeneous Coordinates

Any point (x, y) is written as $(x, y, 1)$. After transformation, divide the first two coordinates by the third (usually 1). This unifies translation with linear transformations.

Example 38. Rotation of a Point: Rotate the point $\mathbf{p} = (3, 1)$ by $90°$ counterclockwise.

Rotation matrix:

$$R = \begin{pmatrix} \cos 90° & -\sin 90° \\ \sin 90° & \cos 90° \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

$$R \begin{pmatrix} 3 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \cdot 3 + (-1) \cdot 1 \\ 1 \cdot 3 + 0 \cdot 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 3 \end{pmatrix}.$$

Result: The point moves from $(3, 1)$ to $(-1, 3)$.

Example 39. Scaling + Rotation (Composite) Scale by $s_x = 2$, $s_y = 0.5$, then rotate by $45°$.

Scaling matrix S:

$$S = \begin{pmatrix} 2 & 0 \\ 0 & 0.5 \end{pmatrix}.$$

Rotation matrix R_{45} ($\cos 45° = \sin 45° = \frac{\sqrt{2}}{2}$):

$$R_{45} = \begin{pmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix}.$$

Composite matrix (apply S first, then R_{45}):

$$M = R_{45}S = \begin{pmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & 0.5 \end{pmatrix} = \begin{pmatrix} \sqrt{2} & -\frac{\sqrt{2}}{4} \\ \sqrt{2} & \frac{\sqrt{2}}{4} \end{pmatrix}.$$

Apply to point $(1, 2)$:

$$M \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} \sqrt{2}(1) - \frac{\sqrt{2}}{4}(2) \\ \sqrt{2}(1) + \frac{\sqrt{2}}{4}(2) \end{pmatrix} = \begin{pmatrix} \frac{\sqrt{2}}{2} \\ \frac{3\sqrt{2}}{2} \end{pmatrix}.$$

Example 40. Translation using Homogeneous Coordinates: Translate $(4, 2)$ by $(3, -1)$.

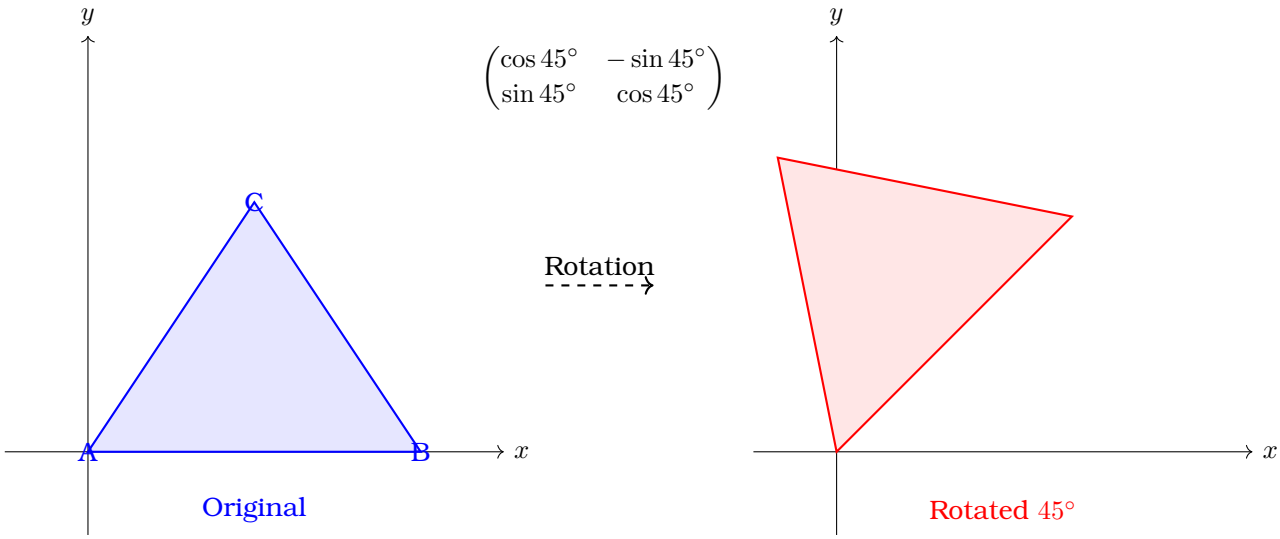
Homogeneous matrix:

$$T = \begin{pmatrix} 1 & 0 & 3 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}.$$

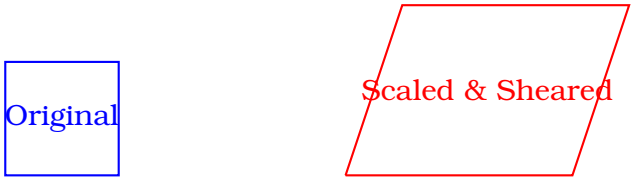
$$T \begin{pmatrix} 4 \\ 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 7 \\ 1 \\ 1 \end{pmatrix} \implies \text{new point } (7, 1).$$

Visualizations

Visualization 1: Rotation of a Triangle

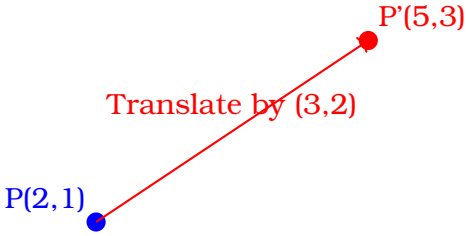


Visualization 2: Scaling + Shearing of a Unit Square



Scaling $S = \begin{pmatrix} 2 & 0 \\ 0 & 1.5 \end{pmatrix}$ Then shear $H = \begin{pmatrix} 1 & 0.5 \\ 0 & 1 \end{pmatrix}$

Visualization 3: Translation in Homogeneous Coordinates



Matrix form: $\begin{pmatrix} 1 & 0 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix}$

12 Real World Applications in Graphics Pipelines

- **Model-View-Projection (MVP) matrix**: Combines model transformation, camera view, and projection into one 4×4 matrix.
- **Animation**: Bone matrices in skeletal animation are linear transformations applied to vertices.
- **Shaders**: GPU runs matrix multiplications thousands of times per frame.
- **Collision detection**: Uses matrix transformations to test rotated bounding boxes.

Example 41. Let T denote the triangle with vertices given by the vectors

$$\mathbf{v}_1 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \quad \mathbf{v}_3 = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$$

as shown in Fig. 1.

a. Stretch the triangle horizontally by a factor of 2.

b. Contract the triangle vertically by a factor of 3.

c. Stretch the triangle horizontally by a factor of 2, and contract the triangle vertically by a factor of 3.

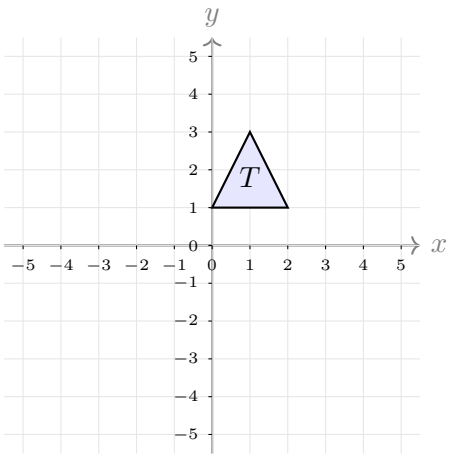


Figure 1

Solution

a. To stretch the triangle horizontally by a factor of 2, we apply the matrix

$$\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$$

to each vertex to obtain

$$\mathbf{v}_1' = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad \mathbf{v}_2' = \begin{pmatrix} 4 \\ 1 \end{pmatrix}, \quad \mathbf{v}_3' = \begin{pmatrix} 2 \\ 3 \end{pmatrix}.$$

Connecting the new vertices by straight-line segments gives the triangle T' shown in Fig. 2(a).

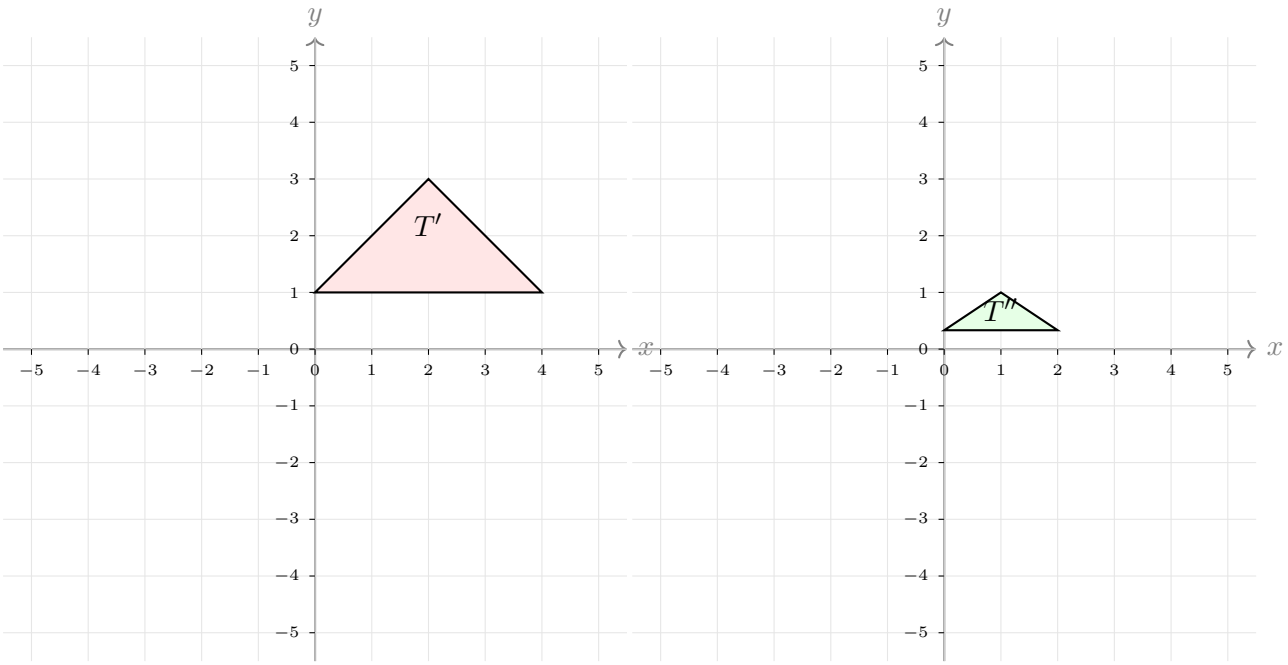
b. To contract the triangle vertically by a factor of 3, we apply the matrix

$$\begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{3} \end{pmatrix}$$

to each vertex to obtain

$$\mathbf{v}_1'' = \begin{pmatrix} 0 \\ \frac{1}{3} \end{pmatrix}, \quad \mathbf{v}_2'' = \begin{pmatrix} 2 \\ \frac{1}{3} \end{pmatrix}, \quad \mathbf{v}_3'' = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

The contracted triangle T'' is shown in Fig. 2(b).



(a) (a) (b) (b)

Figure 2

c. This operator is the composition of the linear operators of parts (a) and (b). The matrix of the operator, relative to the standard basis for $\mathbb{R}^2$, is given by the product

$$\begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{3} \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 0 & \frac{1}{3} \end{pmatrix}.$$

Applying this matrix to the vertices of the original triangle gives

$$\mathbf{v}_1''' = \begin{pmatrix} 0 \\ \frac{1}{3} \end{pmatrix}, \quad \mathbf{v}_2''' = \begin{pmatrix} 4 \\ \frac{1}{3} \end{pmatrix}, \quad \mathbf{v}_3''' = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

as shown in Fig. 3.

Example 42. Perform a vertical shear, with $k = 2$, on the triangle of Fig. 2.

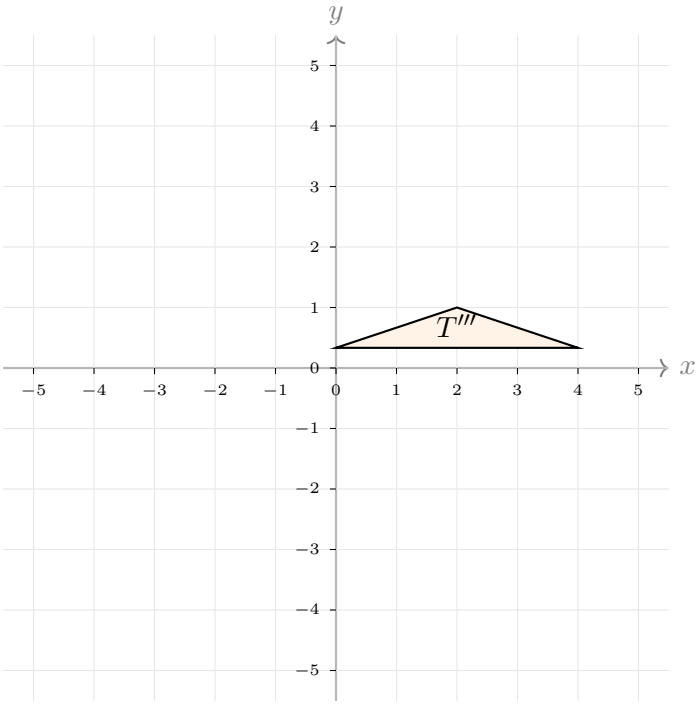


Figure 3

Solution

The matrix of this operator, relative to the standard basis for $\mathbb{R}^2$, is given by

$$\begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}.$$

Applying this matrix to the vertices

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \quad \mathbf{v}_3 = \begin{pmatrix} 3/2 \\ 3 \end{pmatrix}$$

we obtain

$$\mathbf{v}_1' = \begin{pmatrix} 1 \\ 1/3 \end{pmatrix}, \quad \mathbf{v}_2' = \begin{pmatrix} 2 \\ 5 \end{pmatrix}, \quad \mathbf{v}_3' = \begin{pmatrix} 3/2 \\ 3/2 \end{pmatrix}.$$

The images of the original triangle and the sheared triangle are shown in Fig. 4.

Reflection

The reflection of a geometric object through a line produces the *mirror image* of the object across the line.

The linear operator that reflects a vector through the x-axis is given by

$$R_x \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ -y \end{pmatrix}.$$

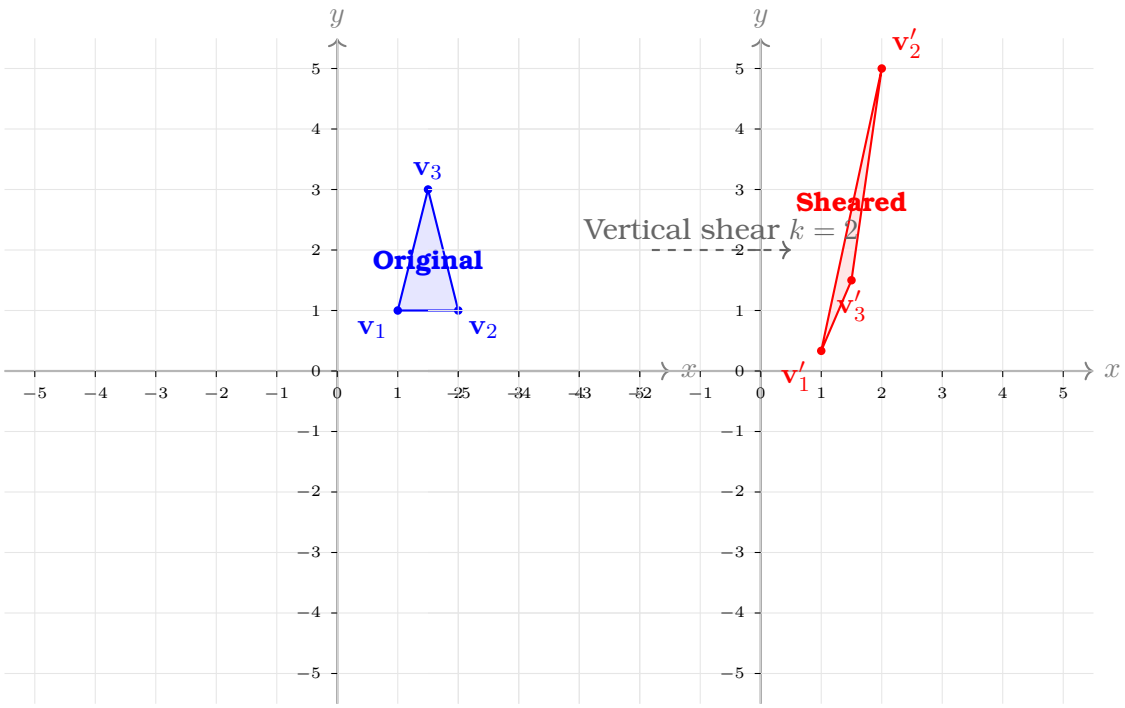


Figure 4

A reflection through the y-axis is given by

$$R_y \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -x \\ y \end{pmatrix}$$

and a reflection through the line $y = x$ is given by

$$R_{y=x} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} y \\ x \end{pmatrix}.$$

The matrix representations, relative to the standard basis B, for each of these are given by

$$[R_x]_B = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad [R_y]_B = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad [R_{y=x}]_B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Example 43. Perform the following reflections on the triangle T of Fig. 1.

a. Reflection through the x-axis.

b. Reflection through the y-axis.

c. Reflection through the line $y = x$.

Solution

a. The vertices of the triangle in Fig. 1 are given by

$$\mathbf{v}_1 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \quad \mathbf{v}_3 = \begin{pmatrix} 1 \\ 3 \end{pmatrix}.$$

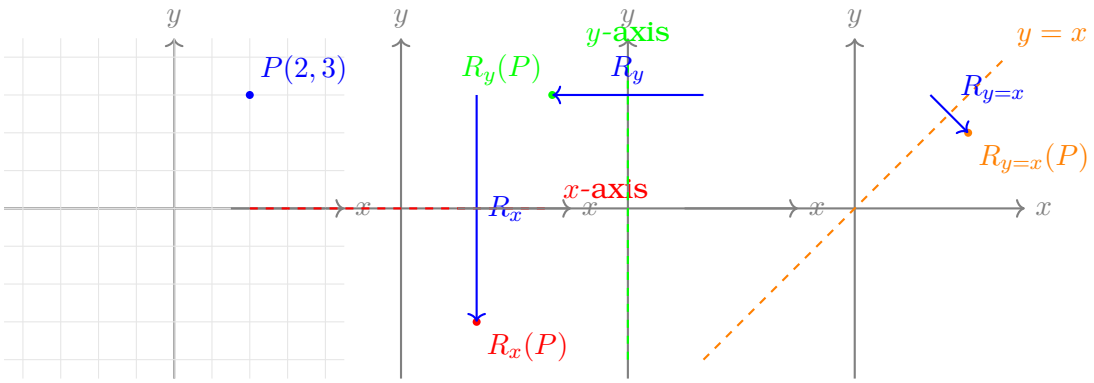


Figure 5: Reflections of point $P(2,3)$ through the x-axis, y-axis, and line $y = x$

Applying the matrix $[R_x]_B$ to the vertices of the original triangle, we obtain

$$\mathbf{v}_1' = \begin{pmatrix} 0 \\ -1 \end{pmatrix}, \quad \mathbf{v}_2' = \begin{pmatrix} 2 \\ -1 \end{pmatrix}, \quad \mathbf{v}_3' = \begin{pmatrix} 1 \\ -3 \end{pmatrix}.$$

The image of the triangle is shown in Fig. 6(a).

b. Applying the matrix $[R_y]_B$ to the vertices of the original triangle, we obtain

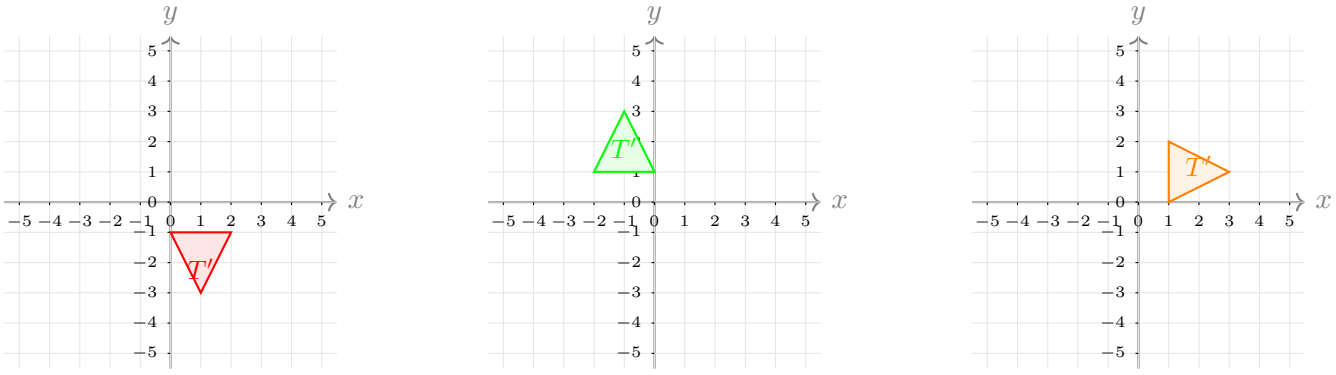
$$\mathbf{v}_1' = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad \mathbf{v}_2' = \begin{pmatrix} -2 \\ 1 \end{pmatrix}, \quad \mathbf{v}_3' = \begin{pmatrix} -1 \\ 3 \end{pmatrix}.$$

The image of the triangle with this reflection is shown in Fig. 6(b).

c. Finally, applying the matrix $[R_{y=x}]_B$ to the vertices of the original triangle, we obtain

$$\mathbf{v}_1' = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \mathbf{v}_2' = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \quad \mathbf{v}_3' = \begin{pmatrix} 3 \\ 1 \end{pmatrix}.$$

The image of the triangle is shown in Fig. 6(c).



(a) (b) (c)

Figure 6

Translation

A translation of a point in the plane moves the point vertically, horizontally, or both. For example, to translate the point $(1, 3)$ three units to the right and two units up, add 3 to the x coordinate and 2 to the y coordinate to obtain the point $(4, 5)$.

Now let $\mathbf{v} = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$ be any vector in $\mathbb{R}^2$ and $\mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}$ some fixed vector. An operation $S : \mathbb{R}^2 \to \mathbb{R}^2$ of the form

$$S(\mathbf{v}) = \mathbf{v} + \mathbf{b} = \begin{pmatrix} v_1 + b_1 \\ v_2 + b_2 \end{pmatrix}$$

is called a translation by the vector $\mathbf{b}$. This transformation is a linear operator if and only if $\mathbf{b} = \mathbf{0}$. Consequently, if $\mathbf{b} \neq \mathbf{0}$, then S cannot be accomplished by means of a 2×2 matrix. However, by using *homogeneous coordinates*, translation of a vector in $\mathbb{R}^2$ can be represented by a 3×3 matrix. The homogeneous coordinates of a vector in $\mathbb{R}^2$ are obtained by adding a third component whose value is 1. Thus, the homogeneous coordinates for the vector $\mathbf{v} = \begin{pmatrix} x \\ y \end{pmatrix}$ are given by

$$\mathbf{w} = \begin{pmatrix} x \\ y \\ 1 \end{pmatrix}.$$

Now, to translate $\mathbf{w}$ by the vector $\mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}$, we let

$$A = \begin{pmatrix} 1 & 0 & b_1 \\ 0 & 1 & b_2 \\ 0 & 0 & 1 \end{pmatrix}$$

so that

$$A\mathbf{w} = \begin{pmatrix} 1 & 0 & b_1 \\ 0 & 1 & b_2 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ 1 \end{pmatrix} = \begin{pmatrix} x + b_1 \\ y + b_2 \\ 1 \end{pmatrix}.$$

To return to $\mathbb{R}^2$, we select the first two components of $A\mathbf{w}$ so that

$$S(\mathbf{v}) = \begin{pmatrix} x + b_1 \\ y + b_2 \end{pmatrix}$$

as desired.

As an illustration of this, let $\mathbf{b} = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$. Using homogeneous coordinates, the 3×3 matrix to perform the translation is

$$A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{pmatrix}.$$

Now let $\mathbf{v} = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$. Then

$$A \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \\ 0 \\ 1 \end{pmatrix}.$$

The vector $S(\mathbf{v}) = \begin{pmatrix} 4 \\ 0 \end{pmatrix}$ is the translation of $\mathbf{v}$ by the vector $\mathbf{b}$.

In the previous illustration the translation can be accomplished with less work by simply adding the vector $\mathbf{b}$ to $\mathbf{v}$. The benefits of using a matrix representation are realized when we combine translation with other types of transformations. To do this, we note that all the previous linear operators can be represented by 3×3 matrices. For example, the 3×3 matrix for reflecting a point (in homogeneous coordinates) through the x-axis is

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Example 44. Find the image of the triangle T of Fig. 1 under a translation by the vector

$$\mathbf{b} = \begin{pmatrix} -5 \\ -3 \end{pmatrix},$$

followed by a horizontal scaling by a factor of 1.5, followed by a reflection through the x-axis.

Solution The matrix for the composition of these operations is given by the product

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1.5 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & -5 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1.5 & 0 & -7.5 \\ 0 & -1 & -3 \\ 0 & 0 & 1 \end{pmatrix}.$$

The vertices of the original triangle in homogeneous coordinates are given by

$$\mathbf{v}_1 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}, \quad \mathbf{v}_3 = \begin{pmatrix} 1 \\ 3 \\ 1 \end{pmatrix}.$$

After applying the above matrix to each of these vectors, we obtain

$$\mathbf{v}_1' = \begin{pmatrix} -7.5 \\ -4 \\ 1 \end{pmatrix}, \quad \mathbf{v}_2' = \begin{pmatrix} -4.5 \\ -4 \\ 1 \end{pmatrix}, \quad \mathbf{v}_3' = \begin{pmatrix} -6 \\ -6 \\ 1 \end{pmatrix}.$$

The resulting triangle, along with the intermediate steps, are shown in Fig. 7.

Example 45. Find a 3×3 matrix that will transform the triangle shown in Fig. 11(a) to the triangle shown in Fig. 8(b).

Solution Triangle T' is obtained from triangle T through a horizontal scaling by a factor of 3, followed by a vertical scaling by a factor of 2, without changing the left vertex $(1, 1)$. The scalings alone will move the point $(1, 1)$ to $(3, 2)$. One way to

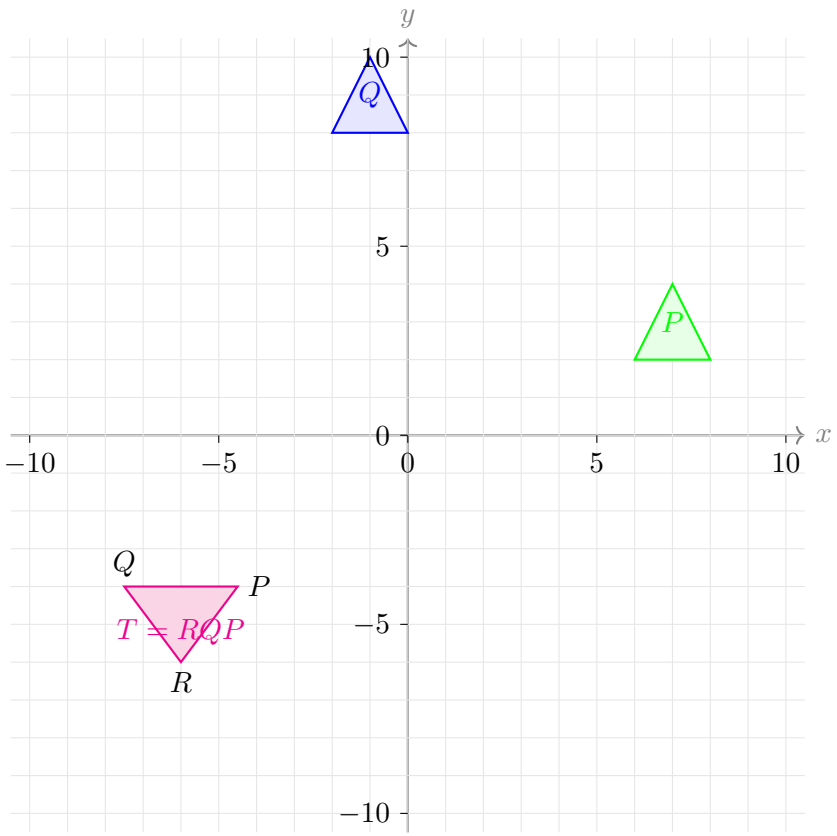


Figure 7: Figure 10

correct this is to first translate the triangle so that the left vertex is located at the origin, perform the scaling, and then translate back. The matrix to perform all these operations is the product of the matrices for each transformation. The matrix is given by

$$\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 3 & 0 & -2 \\ 0 & 2 & -1 \\ 0 & 0 & 1 \end{pmatrix}.$$

Notice that

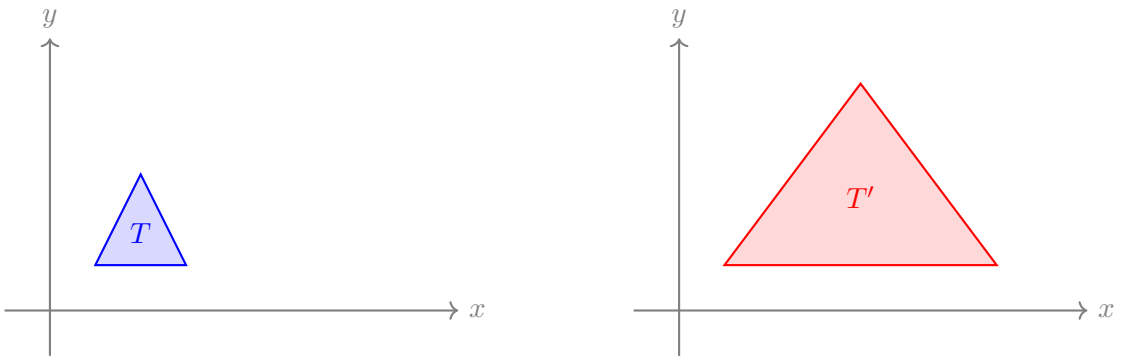
$$\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}^{-1}$$

that is, the matrix representation for translation by $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ is the inverse of the matrix representation for translation by $\begin{pmatrix} -1 \\ -1 \end{pmatrix}$.

Rotation

Another common graphics operation is a rotation through an angle θ.

To describe how a point is rotated, let (x, y) be the coordinates of a point in $\mathbb{R}^2$ and θ a real number. From trigonometry, the new coordinates (x', y') of a point (x, y)



(a) Triangle T (b) Triangle T'

Figure 8

rotated by θ rad about the origin are given by

$$x' = x\cos\theta - y\sin\theta, \quad y' = x\sin\theta + y\cos\theta.$$

If $\theta > 0$, then $\mathbf{v}$ is revolved about the origin in a counterclockwise direction. If $\theta < 0$, the direction is clockwise. These equations define a linear operator $S_\theta : \mathbb{R}^2 \to \mathbb{R}^2$ given by

$$S_\theta \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x\cos\theta - y\sin\theta \\ x\sin\theta + y\cos\theta \end{pmatrix}.$$

The matrix of S_θ relative to the standard basis $B = \{\mathbf{e}_1, \mathbf{e}_2\}$ for $\mathbb{R}^2$ is given by

$$[S_\theta]_B = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}.$$

When using homogeneous coordinates, we apply the matrix

$$\begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Example 46. Find the image of the triangle of Fig. 1 under a translation by the vector $\mathbf{b} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$, followed by a rotation of $30°$, or $\pi/6$ rad, in the counterclockwise direction.

Solution The matrix for the combined operations is given by

$$\begin{pmatrix} \cos\frac{\pi}{6} & -\sin\frac{\pi}{6} & 0 \\ \sin\frac{\pi}{6} & \cos\frac{\pi}{6} & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} & 0 \\ \frac{1}{2} & \frac{\sqrt{3}}{2} & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} & \frac{\sqrt{3}-1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} & \frac{-1-\sqrt{3}}{2} \\ 0 & 0 & 1 \end{pmatrix}.$$

The vertices of the triangle in homogeneous coordinates are given by

$$\mathbf{v}_1 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}, \quad \mathbf{v}_3 = \begin{pmatrix} 1 \\ 3 \\ 1 \end{pmatrix}.$$

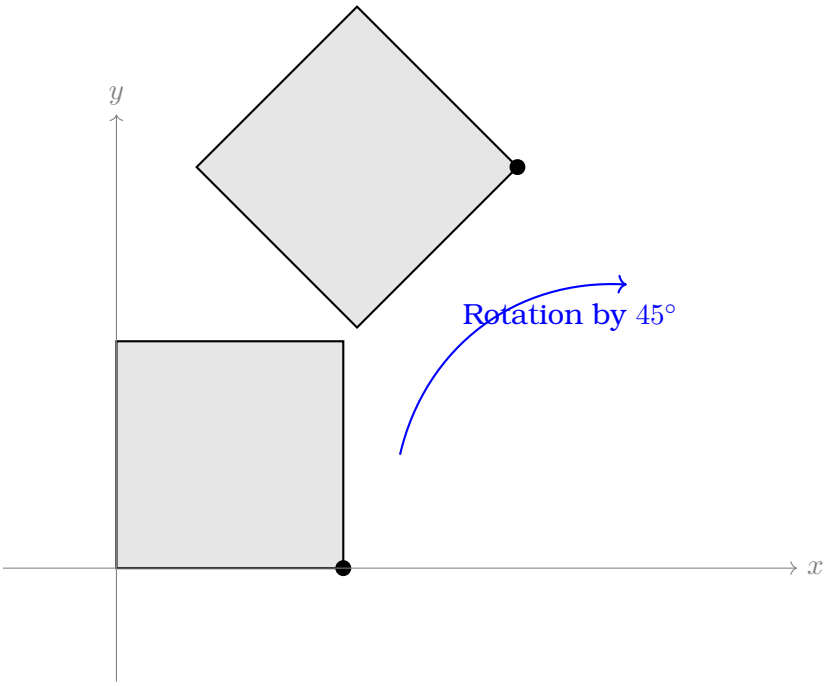


Figure 9

After applying the above matrix to each of these vectors, we obtain

$$\mathbf{v}_1' = \begin{pmatrix} \frac{\sqrt{3}}{2} \\ \frac{1}{2} \\ 1 \end{pmatrix}, \quad \mathbf{v}_2' = \begin{pmatrix} \frac{3\sqrt{3}}{2} \\ \frac{3}{2} \\ 1 \end{pmatrix}, \quad \mathbf{v}_3' = \begin{pmatrix} \sqrt{3}-1 \\ \sqrt{3}+1 \\ 1 \end{pmatrix}.$$

The resulting triangle is shown in Fig. 10.

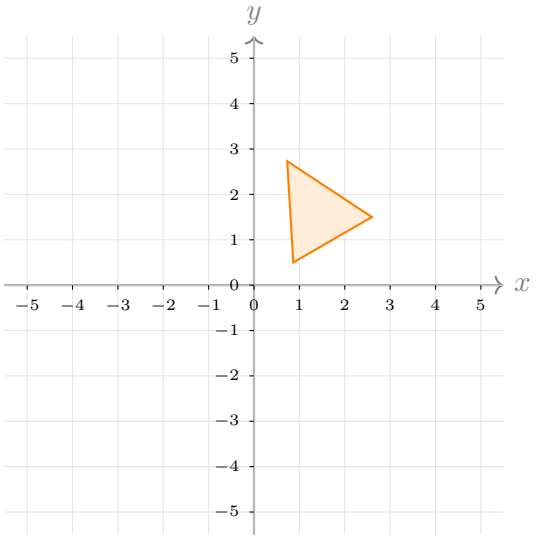


Figure 10

Exercise

Linear transformation

1. Determine whether the function is a linear transformation between vector spaces.

   (a) $T : \mathbb{R}^3 \to \mathbb{R}^3$,
   $$T\left(\begin{bmatrix} x \\ y \\ z \end{bmatrix}\right) = \begin{bmatrix} x + y - z \\ 2xy \\ x + z + 1 \end{bmatrix}$$

   (b) $T : \mathbb{R}^3 \to \mathbb{R}^3$,
   $$T\left(\begin{bmatrix} x \\ y \\ z \end{bmatrix}\right) = \begin{bmatrix} \cos x \\ \sin y \\ \sin x + \sin z \end{bmatrix}$$

   (c) $T : P_3 \to P_3$,
   $$T(p(x)) = 2p''(x) - 3p'(x) + p(x)$$

2. A function $T : V \to W$ between vector spaces and two vectors $\mathbf{u}$ and $\mathbf{v}$ in V are given.

   (i) Find $T(\mathbf{u})$ and $T(\mathbf{v})$.

   (ii) Is $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$?

   (iii) Is T a linear transformation?

   (a) Define $T : P_2 \to P_2$ by
   $$T(p(x)) = p''(x) - 2p'(x) + p(x)$$
   Let
   $$u = x^2 - 3x + 1, \quad v = -x - 1$$

   (b) Define $T : P_3 \to \mathbb{R}^2$ by
   $$T(ax^3 + bx^2 + cx + d) = \begin{bmatrix} -a - b + 1 \\ c + d \end{bmatrix}$$
   Let
   $$u = -x^3 + 2x^2 - x + 1, \quad v = x^2 - 1$$

   (c) Define $T : \mathbb{R}^3 \to \mathbb{R}^2$ by
   $$T\left(\begin{bmatrix} x \\ y \\ z \end{bmatrix}\right) = \begin{bmatrix} x^2 - 1 \\ y + z \end{bmatrix}$$

   (d) Let
   $$\mathbf{u} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} -\frac{1}{2} \\ -1 \\ 1 \end{bmatrix}$$

Range and kernel of linear transformations

3. Define a linear operator $T : \mathbb{R}^3 \to \mathbb{R}^3$ by $T\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x + 2z \\ 2x + y + 3z \\ x - y + 3z \end{pmatrix}$.

   Determine whether the vector $\mathbf{v}$ is in $R(T)$.

   (a) $\mathbf{v} = \begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix}$ (b) $\mathbf{v} = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}$

4. Define a linear transformation $T : M_{2\times 2} \to M_{3\times 2}$ by

$$T\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a + c & b + d \\ -a + 2c & -b + 2d \\ 2a & 2b \end{bmatrix}.$$

   Determine whether the matrix A is in $R(T)$.

   (a) $A = \begin{bmatrix} -1 & -1 \\ -5 & -2 \\ 2 & 0 \end{bmatrix}$ (b) $A = \begin{bmatrix} 1 & 2 \\ 3 & -3 \\ -2 & 2 \end{bmatrix}$

5. Find a basis for the null space of the linear transformation T, $T : \mathbb{R}^2 \to \mathbb{R}^2$,

   (a) $T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 3x + y \\ y \end{pmatrix}$. (b) $T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -x + y \\ x - y \end{pmatrix}$.

6. Find a basis for the range of the linear transformation T.

   (a) $T : \mathbb{R}^3 \to \mathbb{R}^3$,

$$T(\mathbf{v}) = \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & -1 \\ 2 & 0 & 1 \end{bmatrix} \mathbf{v}.$$

   (b) $T : \mathbb{R}^5 \to \mathbb{R}^3$,

$$T(\mathbf{v}) = \begin{bmatrix} 1 & -2 & -3 & 1 & 5 \\ 3 & -1 & 1 & 0 & 4 \\ 1 & 1 & 3 & 1 & 2 \end{bmatrix} \mathbf{v}.$$

Change of basis

7. Find the coordinates of the vector v relative to the ordered basis B.

   (a) $B = \left\{ \begin{bmatrix} 3 \\ 1 \end{bmatrix}, \begin{bmatrix} -2 \\ 2 \end{bmatrix} \right\}$, $v = \begin{bmatrix} 8 \\ 0 \end{bmatrix}$

(b) $B = \left\{ \begin{bmatrix} -2 \\ 4 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \end{bmatrix} \right\}, \quad v = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$

(c) $B = \left\{ \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ -1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} \right\}, \quad v = \begin{bmatrix} 0 \\ 1 \\ \frac{1}{2} \end{bmatrix}$

(d) $B = \{1, x-1, x^2\}, \quad v = p(x) = -2x^2 + 2x + 3$

(e) $B = \{x^2 + 2x + 2, 2x + 3, -x^2 + x + 1\}, \quad v = p(x) = -3x^2 + 6x + 8$

(f) $B = \left\{ \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \right\}, \quad v = \begin{bmatrix} 1 & 3 \\ -2 & 2 \end{bmatrix}$

8. Find the coordinates of the vector v relative to the two ordered bases B_1 and B_2.

(a) $B_1 = \left\{ \begin{bmatrix} -3 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 2 \end{bmatrix} \right\}, \quad B_2 = \left\{ \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}, \quad v = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

(b) $B_1 = \left\{ \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \right\}, \quad B_2 = \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \\ -1 \end{bmatrix} \right\}, \quad v = \begin{bmatrix} -2 \\ 2 \\ 2 \end{bmatrix}$

(c) $B_1 = \{x^2 - x + 1, x^2 + x + 1, 2x^2\}, \quad B_2 = \{2x^2 + 1, -x^2 + x + 2, x + 3\}, \quad v = p(x) = x^2 + x + 3$

(d) $B_1 = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}, \begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix} \right\}, \quad B_2 = \left\{ \begin{bmatrix} 3 & -1 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} -1 & 0 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & -1 \\ 1 & 1 \end{bmatrix} \right\}, \quad v = \begin{bmatrix} 0 & 0 \\ 3 & 1 \end{bmatrix}$

9. Find the transition matrix between the ordered bases B_1 and B_2; then given $[v]_{B_1}$, find $[v]_{B_2}$.

(a) $B_1 = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \end{bmatrix} \right\}, \quad B_2 = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}, \quad [v]_{B_1} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$

(b) $B_1 = \left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\}, \quad B_2 = \left\{ \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ -1 \end{bmatrix} \right\}, \quad [v]_{B_1} = \begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix}$

(c) $B_1 = \{1, x, x^2\}, \quad B_2 = \{x^2, 1, x\}, \quad [v]_{B_1} = \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}$

(d) $B_1 = \{x^2 - 1, 2x^2 + x + 1, -x + 1\}, \quad B_2 = \{(x-1)^2, x + 2, (x+1)^2\}, \quad [v]_{B_1} = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$

(e) Let

$$B_1 = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

be the standard ordered basis for $\mathbb{R}^4$ and let

$$B_2 = \left\{ \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

be a second ordered basis.

a. Find the transition matrix from the ordered basis B_1 to the ordered basis B_2.

b. Find the coordinates of the vector $v = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$ relative to the ordered basis B_2.

Matrix representation and similarity

10. Let $T : \mathbb{R}^n \to \mathbb{R}^n$ is a linear operator.

- Find the matrix representation for T relative to the standard basis for $\mathbb{R}^n$.
- Find $T(\mathbf{v})$, using a direct computation and using the matrix representation.

(a) $T : \mathbb{R}^2 \to \mathbb{R}^2$

$$T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 5x - y \\ -x + y \end{pmatrix}, \qquad \mathbf{v} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

(b) $T : \mathbb{R}^2 \to \mathbb{R}^2$

$$T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -x \\ y \end{pmatrix}, \qquad \mathbf{v} = \begin{pmatrix} -1 \\ 3 \end{pmatrix}$$

(c) $T : \mathbb{R}^3 \to \mathbb{R}^3$

$$T\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -x + y + 2z \\ 3y + z \\ x - z \end{pmatrix}, \qquad \mathbf{v} = \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix}$$

(d) $T : \mathbb{R}^3 \to \mathbb{R}^3$

$$T\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x \\ y \\ -z \end{pmatrix}, \qquad \mathbf{v} = \begin{pmatrix} 2 \\ -5 \\ 1 \end{pmatrix}$$

11. Let $T : V \to V$ is a linear operator with B and B' ordered bases for V.

- Find the matrix representation for T relative to the ordered bases B and B'.

- Find $T(\mathbf{v})$, using a direct computation and using the matrix representation.

(a) $T : \mathbb{R}^2 \to \mathbb{R}^2$

$$T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -x + 2y \\ 3x \end{pmatrix}$$

$$B = \left\{ \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \begin{pmatrix} 2 \\ 0 \end{pmatrix} \right\}, \qquad B' = \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}$$

$$\mathbf{v} = \begin{pmatrix} -1 \\ -2 \end{pmatrix}$$

(b) $T : \mathbb{R}^3 \to \mathbb{R}^3$

$$T\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 2x - z \\ -x + y + z \\ 2z \end{pmatrix}$$

$$B = \left\{ \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} \right\}, B' = \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\}$$

$$\mathbf{v} = \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$$

(c) $T : \mathbb{R}^2 \to \mathbb{R}^2$

$$T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2x \\ x + y \end{pmatrix}$$

$$B = \left\{ \begin{pmatrix} -1 \\ -2 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\}, \quad B' = \left\{ \begin{pmatrix} 3 \\ -2 \end{pmatrix}, \begin{pmatrix} 0 \\ -2 \end{pmatrix} \right\}$$

$$\mathbf{v} = \begin{pmatrix} -1 \\ -3 \end{pmatrix}$$

(d) $T : \mathbb{R}^3 \to \mathbb{R}^3$

$$T\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x + z \\ 2y - x \\ y + z \end{pmatrix}$$

$$B = \left\{ \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \right\}, B' = \left\{ \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \begin{pmatrix} -1 \\ -1 \\ 0 \end{pmatrix} \right\}$$

$$\mathbf{v} = \begin{pmatrix} -2 \\ 1 \\ 3 \end{pmatrix}$$

12. Consider $[T]_{B_1}$ is the matrix representation of a linear operator relative to the basis B_1, and $[T]_{B_2}$ is the matrix representation of the same operator relative to the basis B_2. Show that the action of the operator on the vector $\mathbf{v}$ is the same whether using $[T]_{B_1}$ or $[T]_{B_2}$.

(a)
$$[T]_{B_1} = \begin{pmatrix} 4 & 1 \\ -2 & 3 \end{pmatrix}, \quad [T]_{B_2} = \begin{pmatrix} 5 & 2 \\ -3 & 1 \end{pmatrix},$$
$$B_1 = \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}, \qquad B_2 = \left\{ \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} -1 \\ 1 \end{pmatrix} \right\}, \qquad \mathbf{v} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}.$$

(b)
$$[T]_{B_1} = \begin{pmatrix} 2 & -1 \\ 0 & 4 \end{pmatrix}, \quad [T]_{B_2} = \begin{pmatrix} 1 & 3 \\ -2 & 2 \end{pmatrix},$$
$$B_1 = \{\mathbf{e}_1, \mathbf{e}_2\}, \qquad B_2 = \left\{ \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ -2 \end{pmatrix} \right\}, \qquad \mathbf{v} = \begin{pmatrix} -1 \\ 3 \end{pmatrix}.$$

13. In the following two questions, a linear operator T and bases B_1 and B_2 are given.

- Find $[T]_{B_1}$ and $[T]_{B_2}$.
- Verify that the action on $\mathbf{v}$ of the linear operator T is the same when using the matrix representation of T relative to the bases B_1 and B_2.

(a)
$$T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2x - y \\ x + 3y \end{pmatrix}, \qquad B_1 = \{\mathbf{e}_1, \mathbf{e}_2\},$$
$$B_2 = \left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}, \qquad \mathbf{v} = \begin{pmatrix} 4 \\ -2 \end{pmatrix}.$$

(b)
$$T\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x + y + z \\ 2x - z \\ y + 2z \end{pmatrix}, \qquad B_1 = \{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\},$$
$$B_2 = \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \right\}, \qquad \mathbf{v} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}.$$

14. In the following two questions, $[T]_{B_1}$ and $[T]_{B_2}$ are the matrix representations of a linear operator relative to the bases B_1 and B_2. Find the transition matrix $P = [I]_{B_2}^{B_1}$ and use the change-of-basis theorem to show directly that the matrices are similar.

(a)
$$[T]_{B_1} = \begin{pmatrix} 3 & 1 \\ -1 & 2 \end{pmatrix}, \quad [T]_{B_2} = \begin{pmatrix} 1 & 2 \\ 0 & -1 \end{pmatrix},$$
$$B_1 = \{\mathbf{e}_1, \mathbf{e}_2\}, \qquad B_2 = \left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\}.$$

(b)

$$[T]_{B_1} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad [T]_{B_2} = \begin{pmatrix} 2 & -1 \\ 1 & 1 \end{pmatrix},$$

$$B_1 = \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}, \qquad B_2 = \left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} -1 \\ 2 \end{pmatrix} \right\}.$$

15. In the following two questions, find the matrix representation of the linear operator T relative to B_1. Then use the change-of-basis theorem to find $[T]_{B_2}$.

(a)

$$T \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 3x - 2y \\ 4x + y \end{pmatrix}, \qquad B_1 = \{\mathbf{e}_1, \mathbf{e}_2\},$$

$$B_2 = \left\{ \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 3 \\ 1 \end{pmatrix} \right\}.$$

(b)

$$T \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x - y + z \\ 2x + 3z \\ y - 2z \end{pmatrix}, \qquad B_1 = \{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\},$$

$$B_2 = \left\{ \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right\}.$$

Additional Exercises

1. Determine whether the function is a linear transformation.

   (a) $T : \mathbb{R}^2 \to \mathbb{R}^2$, $T(x, y) = (x, 1)$

   (b) $T : \mathbb{R}^2 \to \mathbb{R}^2$, $T(x, y) = (x, y^2)$

   (c) $T : \mathbb{R}^3 \to \mathbb{R}^3$, $T(x, y, z) = (x + y, x - y, z)$

   (d) $T : \mathbb{R}^2 \to \mathbb{R}^3$, $T(x, y) = (\sqrt{x}, xy, \sqrt{y})$

   (e) $T : \mathbb{R}^3 \to \mathbb{R}^3$, $T(x, y, z) = (x^2, xy, y^2)$

2. For the linear transformations $T : \mathbb{R}^3 \to \mathbb{R}^3$, determine $T(x, y, z)$.

   (a) $T(2, 0, 2) = (2, 4, -1)$, $T(4, 2, 0) = (1, 3, -2)$, $T(0, 2, 5) = (0, -2, 2)$

   (b) $T(1, 1, 1) = (2, 0, -1)$, $T(0, -1, 2) = (-3, 2, -1)$, $T(1, 0, 1) = (1, 1, 0)$

3. Find the linear transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$ given by

$$A = \begin{pmatrix} a & -b \\ b & a \end{pmatrix}$$

   such that $T(12, 5) = (13, 0)$.

4. Define the linear transformation T by $T(\mathbf{v}) = A\mathbf{v}$. Find (i) $\ker(T)$ or $\text{null}(T)$, (ii) $\text{nullity}(T)$, (iii) $\text{range}(T)$ or $\text{Im}(T)$, and (iv) $\text{rank}(T)$.

   (a) $A = \begin{pmatrix} 2 & 0 \\ 2 & 4 \end{pmatrix}$

   (b) $A = \begin{pmatrix} 5 & 1 \\ -3 & 1 \\ 1 & -1 \end{pmatrix}$

   (c) $A = \begin{pmatrix} \frac{4}{9} & -\frac{4}{9} & \frac{2}{9} \\ \frac{4}{9} & \frac{4}{9} & -\frac{2}{9} \\ \frac{2}{9} & -\frac{2}{9} & -\frac{1}{9} \end{pmatrix}$

   (d) $A = \begin{pmatrix} 2 & 2 & -3 & 1 \\ 3 & 3 & -5 & 0 \\ 2 & -3 & 4 & -4 \end{pmatrix}$

5. If $T : \mathbb{R}^3 \to \mathbb{R}^3$ is defined by $T(x, y, z) = (2x - z, 3x - 2y, x - 2y + z)$,

   (a) Determine the null space and nullity of T

   (b) Determine whether T is one-one and onto

   (c) Find a basis for null space of T

   (d) Determine the range space or image space and rank of T

   (e) Find a basis of range space of T

6. Determine whether the given linear transformations are invertible or not. Find a formula for inverse T^{-1}, if invertible.

   (a) $T(x, y, z) = (3x, x - y, 2x + y + z)$

   (b) $T(x, y, z) = (2x, 4x - y, 2x + 3y - z)$

7. Let T be the linear operator on $\mathbb{R}^3$ defined by

$$T(x_1, x_2, x_3) = (3x_1 + x_2, -2x_1 + x_2, -x_1 + 2x_2 + 4x_3).$$

   What is the matrix of T in the ordered basis $\{\alpha_1, \alpha_2, \alpha_3\}$ where

$$\alpha_1 = (1, 0, 1), \quad \alpha_2 = (-1, 2, 1), \quad \alpha_3 = (2, 1, 1)?$$

8. Find the matrix representation of each of the following linear transformations T on $\mathbb{R}^2$ w.r.t the standard basis $\{e_1, e_2\}$.

   (a) $T(x, y) = (2y, 3x - y)$

   (b) $T(x, y) = (3x - 4y, x + 5y)$

9. Consider the following ordered bases of $\mathbb{R}^3$: $B_1 = \{e_1, e_2, e_3\}$ the standard basis and

$$B_2 = \{(1, 1, 1), (1, 1, 0), (1, 0, 0)\}.$$

(a) Find the transition matrix P from B_1 to B_2.

(b) Find the transition matrix Q from B_2 to B_1.

(c) Show that $[\mathbf{v}]_{B_2} = P[\mathbf{v}]_{B_1}$ for any vector $\mathbf{v} \in \mathbb{R}^3$.

10. Let $T : \mathbb{R}^3 \to \mathbb{R}^2$ be the LT defined by $T(x, y, z) = (3x + 2y - 4z, x - 5y + 3z)$, and let $B_1 = \{(1, 1, 1), (1, 1, 0), (1, 0, 0)\}$ and $B_2 = \{(1, 3), (2, 5)\}$ be bases for $\mathbb{R}^3$ and $\mathbb{R}^2$, respectively.

(a) Find $[T]_{B_2}^{B_1}$.

(b) Verify $[T]_{B_2}^{B_1}[\mathbf{v}]_{B_1} = [T(\mathbf{v})]_{B_2}$ for any $\mathbf{v} \in \mathbb{R}^3$.

11. Consider the linear mapping $T : \mathbb{R}^2 \to \mathbb{R}^2$ defined by $T(x, y) = (3x + 4y, 2x - 5y)$ and the following basis of $\mathbb{R}^2$: $E = \{e_1, e_2\} = \{(1, 0), (0, 1)\}$ and $S = \{u_1, u_2\} = \{(1, 2), (2, 3)\}$.

(a) Find the matrix A representing T relative to E.

(b) Find the matrix B representing T relative to S.

12. Find the transition matrix between the ordered basis

$$B_1 = \{x^2 - 1,\ 2x^2 + x + 1,\ -x + 1\}$$

and

$$B_2 = \{(x - 1)^2,\ x + 2,\ (x + 1)^2\}.$$

If $[\mathbf{v}]_{B_1} = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$, find $[\mathbf{v}]_{B_2}$.

13. Let $T : \mathbb{R}^3 \to \mathbb{R}^3$ be a linear operator and $B = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ be a basis for $\mathbb{R}^3$. Suppose

$$T(\mathbf{v}_1) = \begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix}, \quad T(\mathbf{v}_2) = \begin{pmatrix} 0 \\ 5 \\ 0 \end{pmatrix}, \quad T(\mathbf{v}_3) = \begin{pmatrix} -1 \\ -1 \\ 2 \end{pmatrix}.$$

(a) Determine whether $\mathbf{w} = \begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix}$ is in the range of T.

(b) Find the basis for $R(T)$.

(c) Find $\dim N(T)$.

14. Define a linear operator $T : P_2 \to P_2$ by $T[p(x)] = p'(x)$,

(a) Define the range of T.

(b) Find $\dim R(T)$.

15. A function is defined as $T : P_3 \to \mathbb{R}^2$, by

$$T(ax^3 + bx^2 + cx + d) = \begin{pmatrix} -a - b + 1 \\ c + d \end{pmatrix}.$$

Let $u = -x^3 + 2x^2 - x + 1$ and $v = x^2 - 1$. Find $T(u)$ and $T(v)$. Is T a linear transformation?

16. Let $T : P_2 \to P_2$ is a linear operator with B and B' ordered basis for V, such that

$$T(ax^2 + bx + c) = ax^2 + bx + c, \quad B = \{1, 1 - x, (1 - x)^2\} \quad \text{and} \quad B' = \{1, x, x^2\}$$

and $v = x^2 - 3x + 3$. Find the matrix representation for T relative to the ordered basis B and B'. Find $T(v)$, using direct computation and using matrix representation.

17. Two bases for a vector space are $B_1 = \{1, x, x^2, x^3\}$ and $B_2 = \{x, x^2, 1, x^3\}$. Find the transition matrix from B_1 to B_2 and from B_2 to B_1.

18. Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be a linear transformation defined by

$$T(x, y) = (x + y, x - y),$$

And let $\alpha = \{(1, 1), (1, 0)\}$ and $\beta = \{(1, 2), (1, -1)\}$ be ordered basis for $\mathbb{R}^2$ and $\mathbb{R}^2$, respectively. Find the associated matrix $[T]_\beta^\alpha$ for T and verify $[T]_\beta^\alpha[\mathbf{v}]_\alpha = [T(\mathbf{v})]_\beta$ for any $\mathbf{v} \in \mathbb{R}^2$.

19. Suppose the mapping $F : \mathbb{R}^2 \to \mathbb{R}^2$ is defined by $F = (x + y, x)$ show that F is linear.

20. Describe explicitly the linear transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$ such that $T(2, 3) = (4, 5)$ and $T(1, 0) = (0, 0)$.

21. Let T be a linear operator on $V_3(\mathbb{R})$ defined by

$$T(a, b, c) = (3a, a - b, 2a + b + c) \ \forall \ (a, b, c) \in V_3(\mathbb{R}).$$

Is T invertible? If so find a rule for T^{-1} like the one which defines T.

22. Let $T : V_4(\mathbb{R}) \to V_3(\mathbb{R})$ be a linear transformation defined by

$$T(a, b, c, d) = (a - b + c + d,\ a + 2c - d,\ a + b + 3c - 3d).$$

Then obtain the basis and dimension of the range space of T and null space of T.

23. Let $T : \mathbb{R}^3 \to \mathbb{R}^4$ be a linear transformation defined by

$$T(x, y, z) = (2x + y + 4z,\ x + y - 2z,\ y + 2z,\ x + y + 3z).$$

Find the image and kernel of T. What is the dimension of $\mathrm{Im}(T)$?